AASTU
UNIVERSITY FOR INDUSTRY

College Engineering
Department of Mechanical Engineering

Engineering Drawing (MEng2101)

Chapter 1: Introduction to Engineering Drawing

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October, 2026

Topics

- History of Drawing
- Objective of Engineering Drawing
- Lettering
- Drawing Tools & Instruments
- Line Types
- Basic Geometric Construction

Learning Outcomes

After completing this chapter, students will be able to:

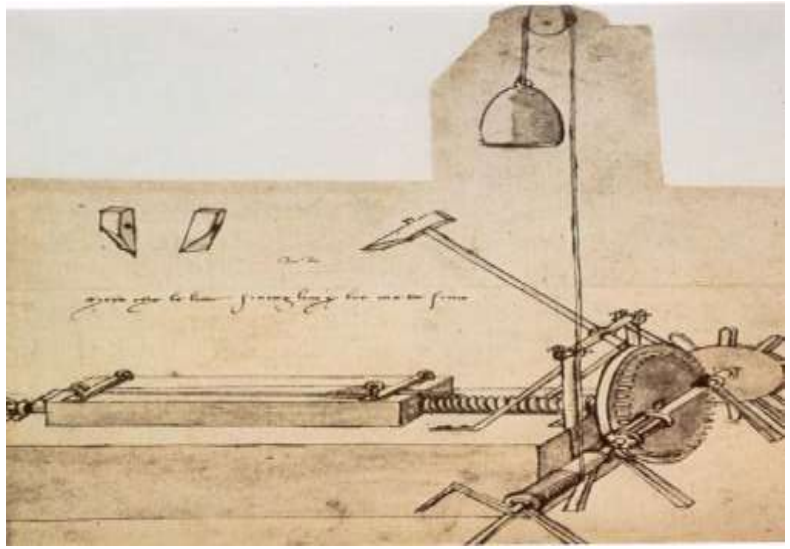
- ✓ Read and interpret basic engineering drawings and their importance in the engineering field.
- ✓ Apply proper lettering techniques for clear and professional engineering drawings.
- ✓ Demonstrate proficiency in using drawing instruments and tools to create precise engineering drawings.
- ✓ Apply the appropriate line types to different elements of an engineering drawing, such as outlines, hidden lines, and centerlines.
- ✓ Apply geometric construction techniques in the creation of accurate engineering drawings.

1.1 History of Engineering Drawing

- For centuries, engineering was focused on war and few documents exist today describing how these early military war machines were built. Those that do exist were scratched into clay tablets.
- During 15th century, the concept of graphic projections was well understood by early Italian architects.
  - ✓ time paper began as drawing medium
- Drawings describing machines and buildings date back to the 14th and 15th centuries described more as sketches than as technical drawings.
  - ✓not to scale nor have dimensions.
  - ✓documents contain extensive textual descriptions

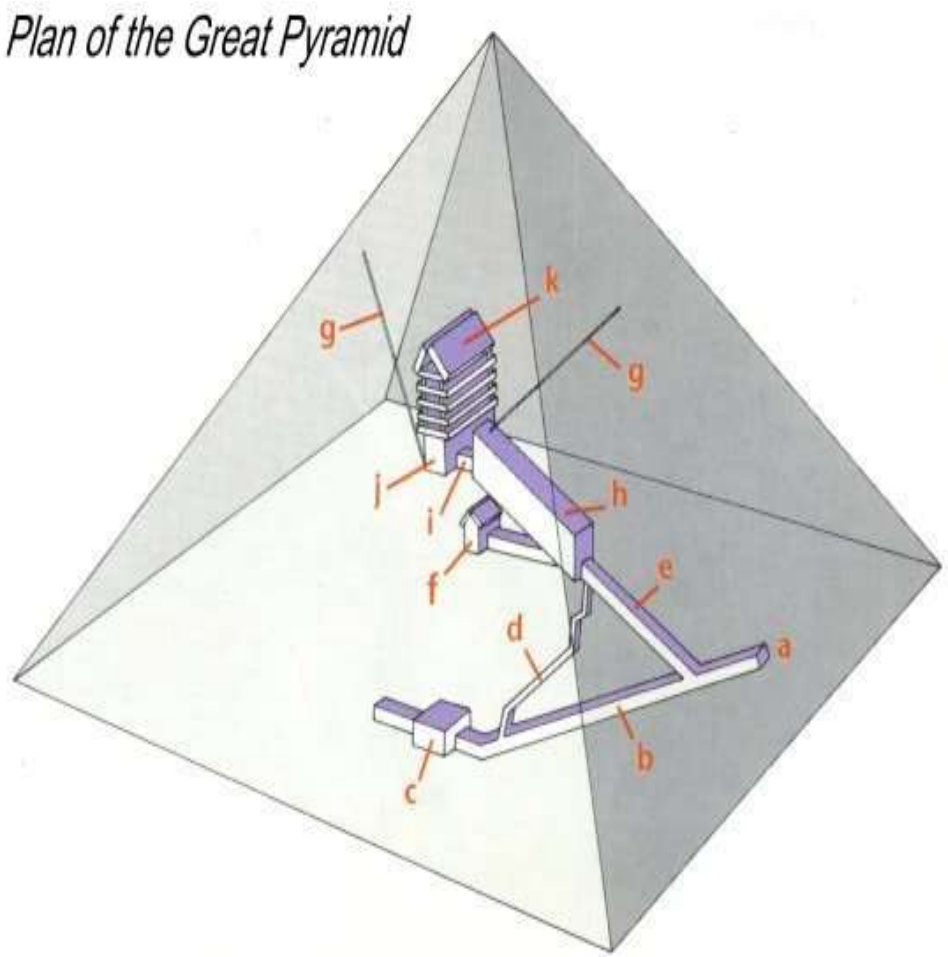
History…

- Some of the early engineering drawings is the work of Leonardo da Vinci.
  - ✓known for his Mona Lisa,
  - ✓designer of military machines and forerunners of today's industrial machines.
  - ✓design work was artistic in nature-more illustration than engineering drawings.
  - ✓no multi-view drawings of any of his designs are known to exist today.



History ….

- Ancient civilizations like Egypt and Mesopotamia used basic pictorial symbols.
- Pyramids and temples were built using rudimentary plans.



AXONOMETRIC VIEW OF THE PYRAMID OF KHUFU

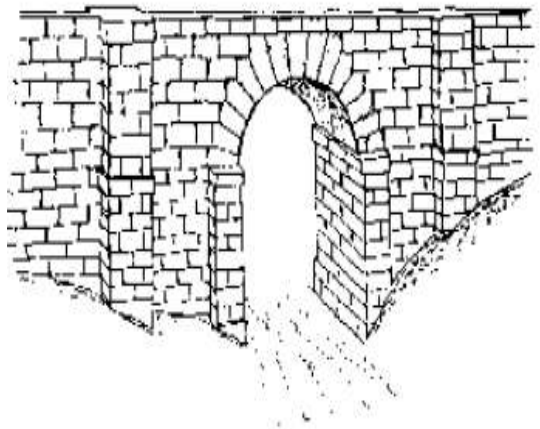
a. Entrance
b. Descending corridor
c. Underground chamber
d. Service corridor
e. Ascending corridor
f. Queen's room
g. Air shafts
h. Great Gallery
i. Antechamber
j. King's chamber
k. Weight relief chambers



tombs of the pharaohs

History ….

- Roman Arch and Roman road still in use today
- Marcus Vitruvius provided written records of Design and Construction



- Roman Arch

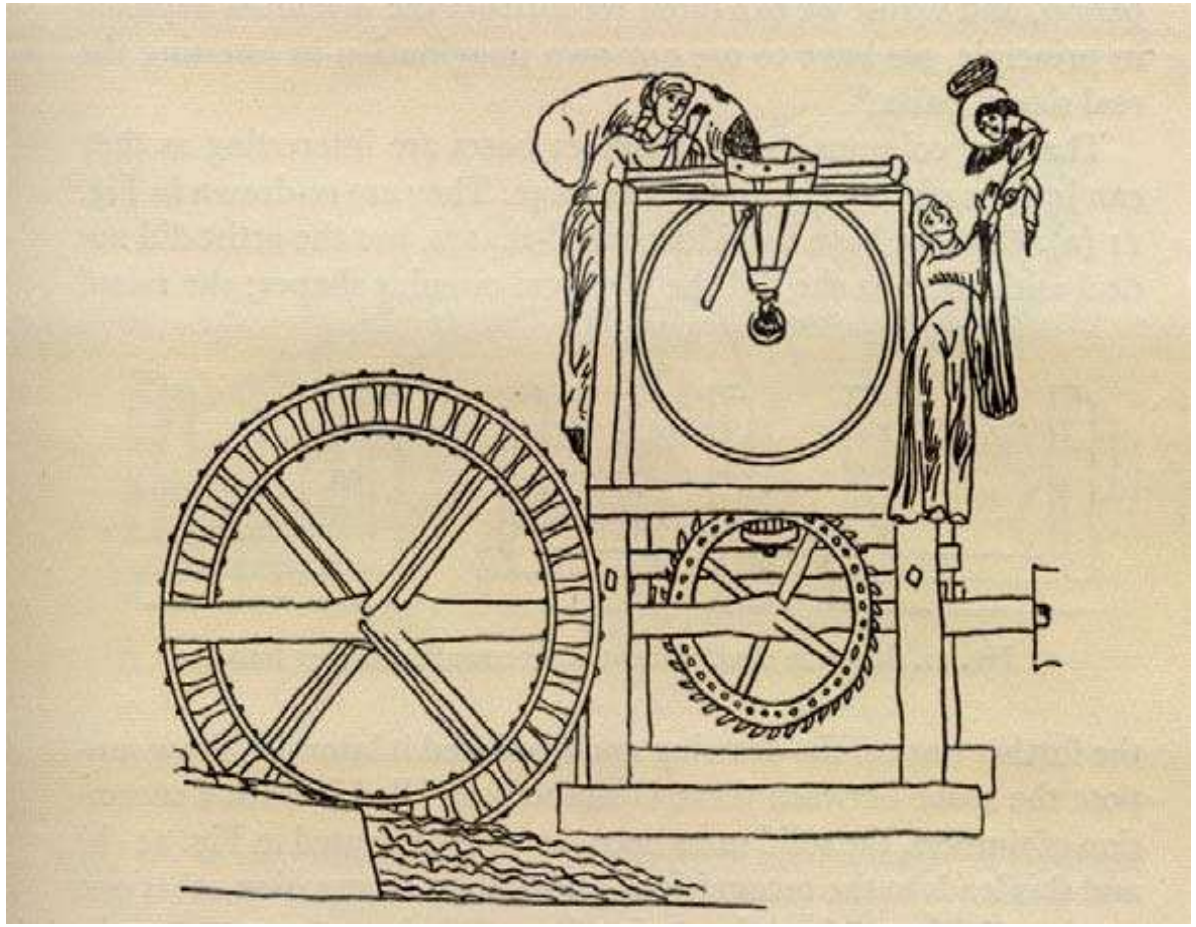


Roman aqueduct, 19 B.C.



Roman Road, near Rome, 300 B.C.

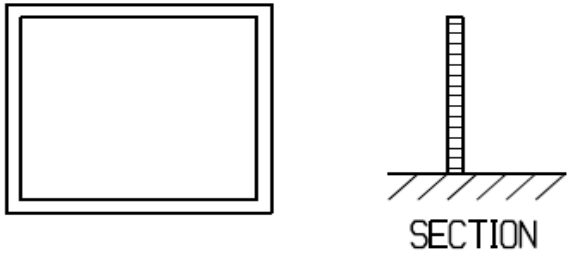
History ….

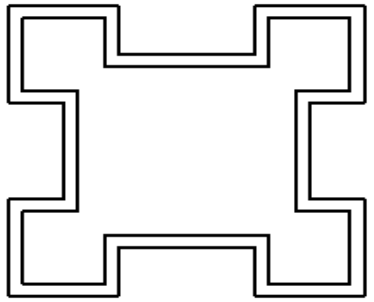


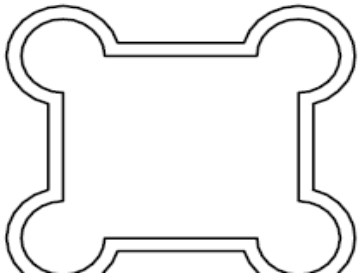
Water Mill

- Perspective projection unknown Flattened and distorted, but records basic principles

History ….







- Design of fortifications
- Simple wall geometries

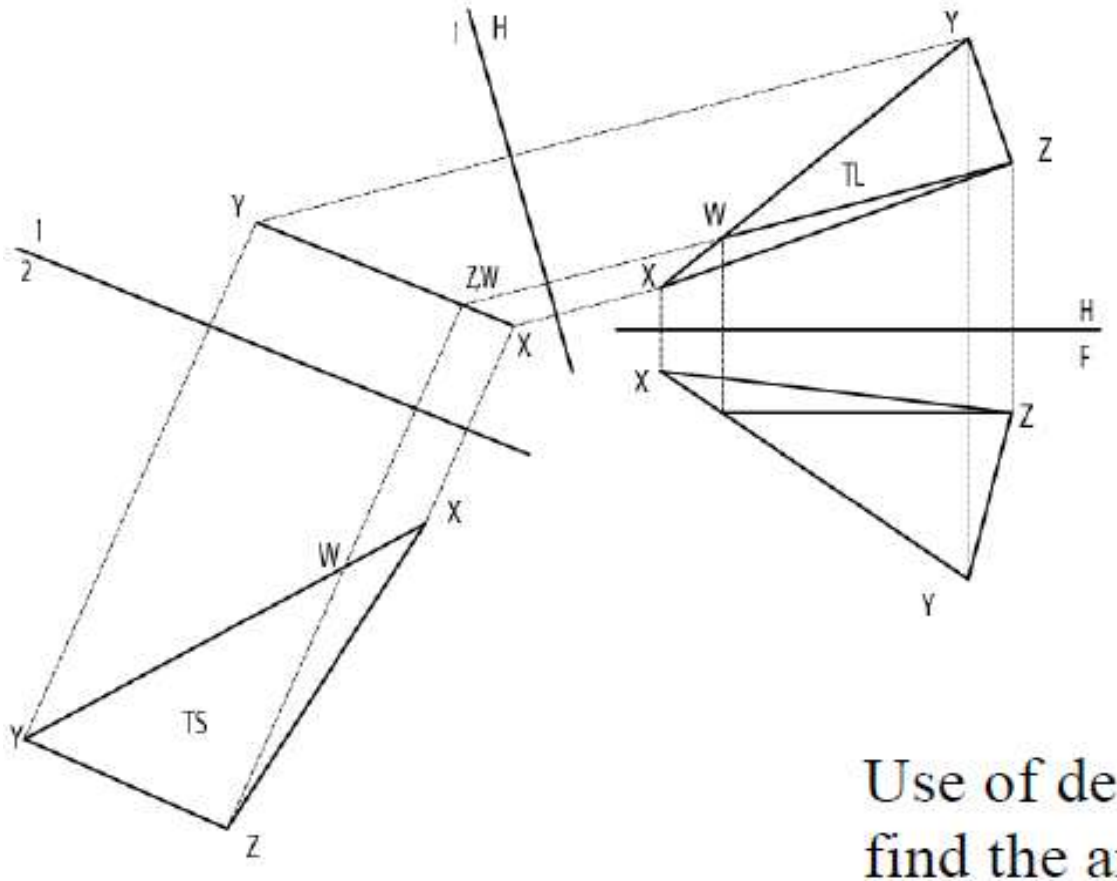




European fortifications

History ….

- Leon Battista Alberti, in 1435 and 1436, proposed drawings with multiple views rather than the single view then common.
- René Descartes (1596–1650) and Gaspard Monge (1746–1818) developed descriptive geometry.



Use of descriptive geometry to find the area of plane XYZ

History ….

- The Industrial Revolution brought about dramatic changes in society, technology, and manufacturing processes. With the rise of mass production and mechanical engineering, there was an increasing need for precise, repeatable, and standardized technical drawings.

  - ✓ 1700's-Appearance of engineering specialization. (Previously all military)
  - ✓ 1750's-Civil Engineering (roads, bridges buildings). Military Engineering (forts, cannon, ships).
  - ✓ 1850's-Mechanical Engineering (steam engines, boilers, machine tools)
  - ✓ 1875-Electrical engineering (motors, generators, power distribution)
  - ✓ 1895 - Chemical Engineering (dyes, solvents, polymers, chemical processes)
  - ✓ 1950's-Nuclear Engineering (reactors, nuclear processes). Industrial Engineering (efficiency, material flow)

History ….

- By the 20th century, engineering drawing had become a formalized discipline.
- Standardization bodies such as the International Organization for Standardization (ISO) and the American National Standards Institute (ANSI) developed universal conventions for line types, symbols, dimensions, and layouts.
- These standards helped eliminate ambiguity and ensure that technical drawings could be universally understood, regardless of language or location.
- During this time, engineering drawing was also fully integrated into engineering education, becoming a core subject for students in technical and engineering fields.

History ….

Tools for Engineering Graphics (Manual tools)



)ld stuff…



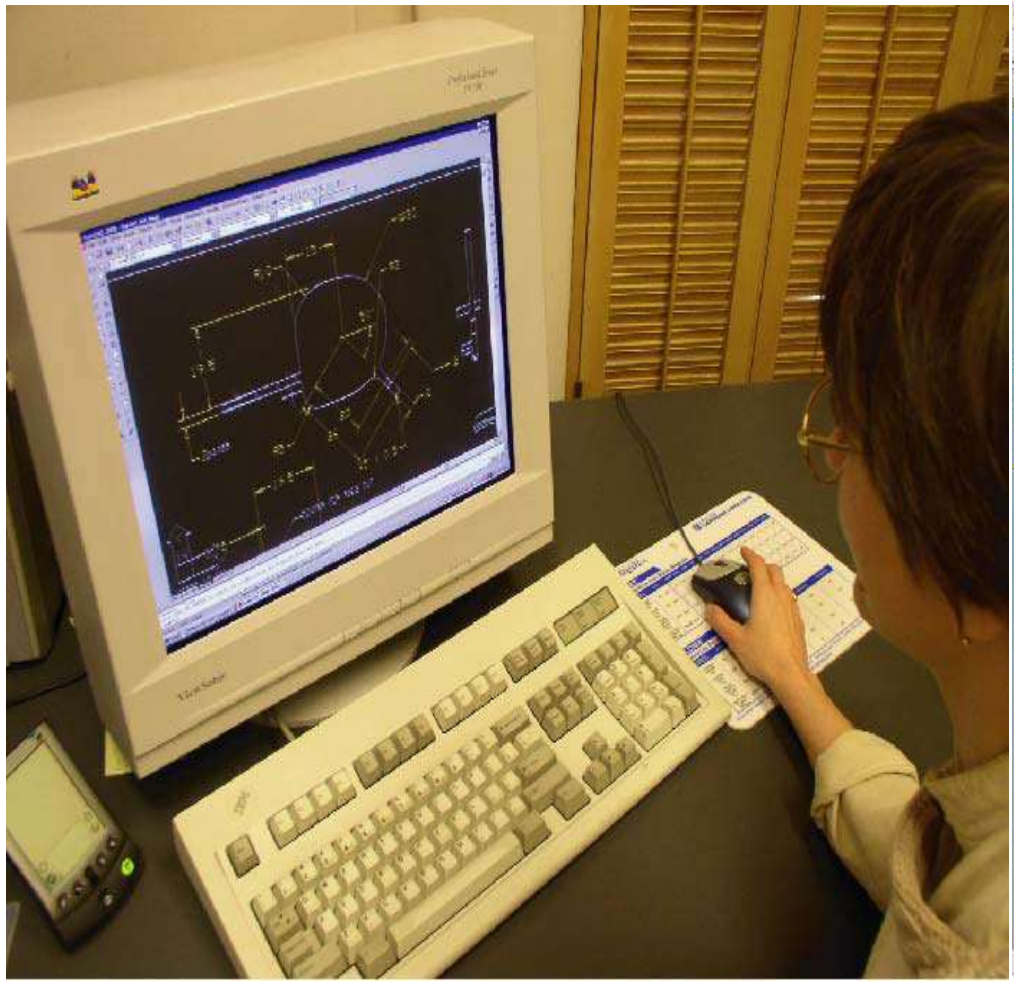


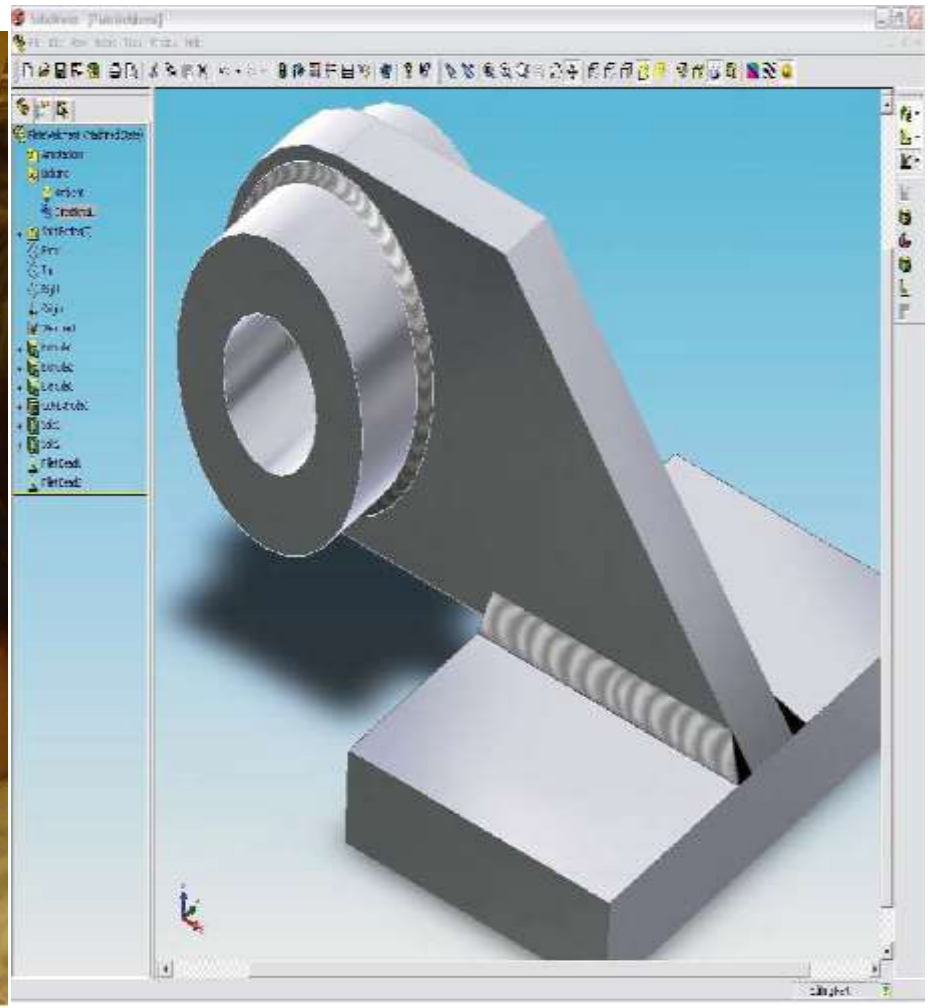
History ….

- The introduction of Computer-Aided Design (CAD) in the 1980s revolutionized engineering drawing.
- CAD software allowed engineers and designers to create, modify, analyze, and optimize designs with far greater speed and precision than manual drafting.
- This shift marked the transition from traditional paper-based drawings to digital formats, enabling 2D drafting as well as complex 3D modeling.
- The digital era continues to evolve with advancements in software, cloud collaboration, and simulation tools, making engineering drawing more dynamic, interactive, and accessible than ever before.

History ….

Tools for Engineering Graphics (CAD tools)





1.2 Objective of Engineering Drawing

- **Drawing** is a graphic language by which communication is accomplished through sketches (drawings).
- If compared with verbal or written description, drawings provide a far better idea about the shape, size and appearance of any object or situation or location, that too fairly quickly.

Effectiveness of Graphics Language



- Try to write a description of an object.
- You can easily understand that the word language are inadequate for describing the size, shape and features completely as well as concisely.

A simple component!

- How hard to define in words?





- Try to describe in words the following representation: A MECHANISM

Composition of Graphic Language

- Graphic language in **engineering application** use ***lines*** to represent the ***surfaces***, ***edges*** and ***contours*** of objects.
  - ✓ The language is known as ***drawing*** or ***drafting*** .
  - ✓ A drawing can be done using ***freehand***, ***instruments*** or ***computer*** methods.

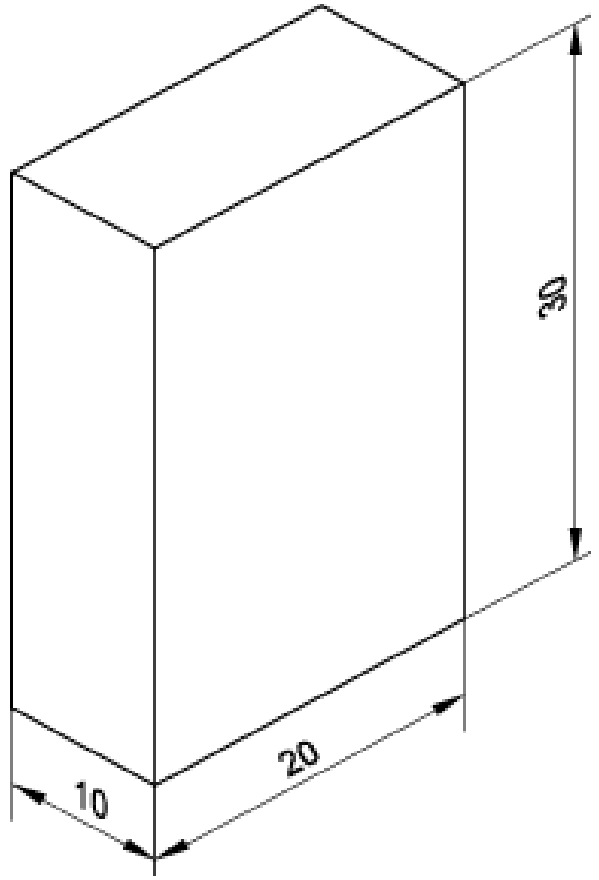
Types of Drawing

There are two basic types of drawings:

- Artistic Drawing



- Technical Drawing



Types of Drawing …

- **Artistic drawing** is used to express the **esthetic idea (beauty), philosophic imaginations** and **abstract ideas** of the artist.
- **Technical Drawing** is the study of the **procedures, tools, skills** and **techniques** used to record and communicate the shape and size of a product. It is a **universal standardized graphical language** used to describe accurately and clearly the form, size and construction of an object. Must contain everything needed for proper interpretation of the design because design and manufacturing may be located far apart - often in different countries.

Types of technical drawing

- Engineering drawing
- Cut away drawing
- Exploded drawing
- Patent drawing
- Technical illustrations
- Technical sketches
- Working drawings, e.g. Plumbering and electrical drawings.

Engineering drawing

- Engineering drawing is a graphical language that communicates ideas and information from one mind to another.
- It is **the language of engineers**.
- Essential for communication across all technical levels - from **engineers to craftsmen**.
- Vital for ensuring **high quality in size, dimension, fit, tolerance, and finish** of products.

Mechanical Engineering drawing:

- Some examples of mechanical engineering drawings are part and assembly drawings, riveted joints, welded joints, fabrication drawings, pneumatic and hydraulic drawings, pipeline diagrams, keys coupling drawings etc.

Electrical and Electronics Engineering drawings

- Wiring diagrams for buildings and industries, circuit diagrams, electrical installation drawings, Circuit drawings, PCB tracks drawings, etc.

Civil Engineering Drawings

- Plan, front elevation of building to be built, foundation drawings etc.

Elements of Engineering Drawing

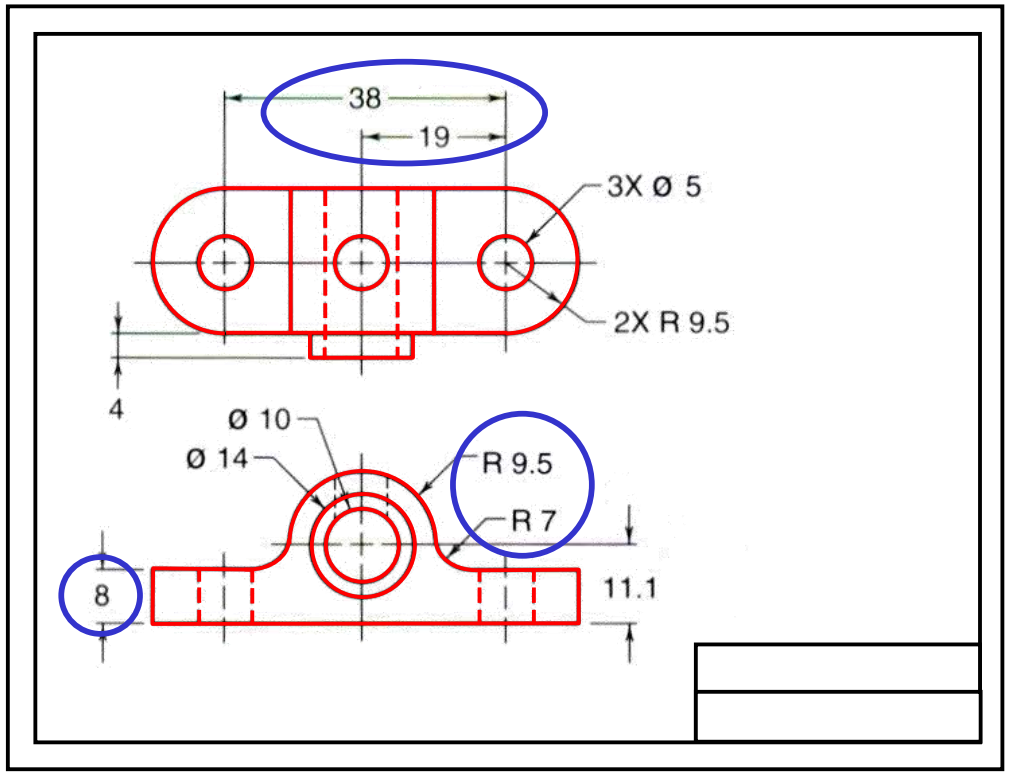
- Engineering drawing are made up of ***graphics language*** and ***word language***.

Graphics language

Describe a shape (mainly).

Word language

Describe size, location and specification of the object.

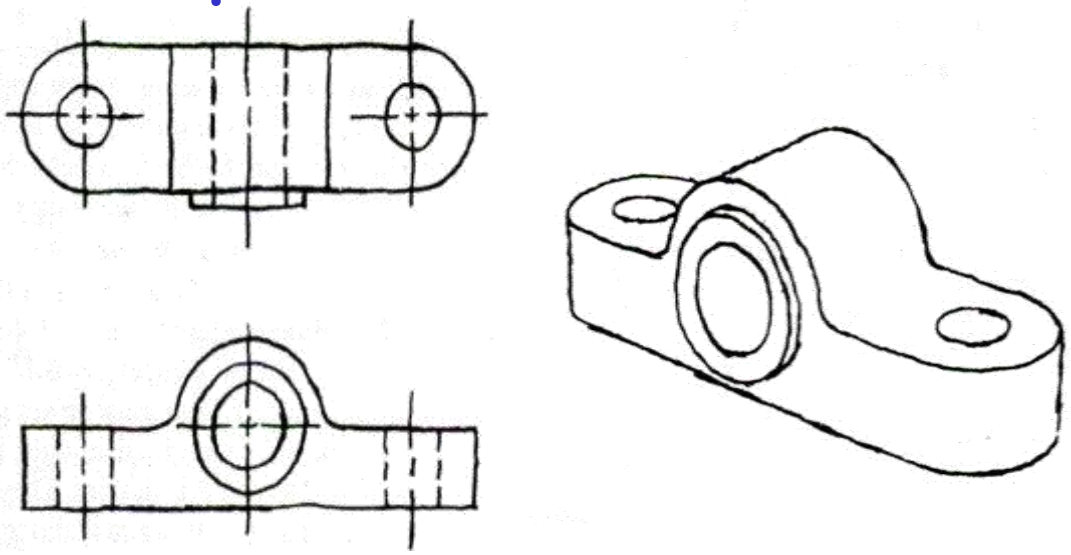


Ways to Create an Engineering Drawing

A drawing can be created in three ways:

1. Freehand sketch: *The lines are sketched without using instruments other than pencils and erasers.*
2. Drawing instruments: *Instruments are used to draw straight lines, circles, and curves concisely and accurately. Thus, the drawings are usually made to scale.*
3. Using a computer: *The drawings are usually made by commercial software such as AutoCAD, solid works, Catia, etc.*

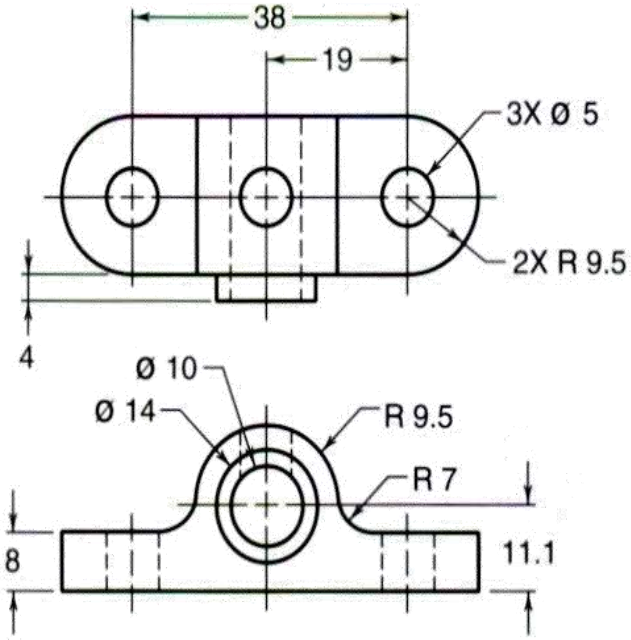
Examples



Freehand drawing



Instrument drawing







Computer drawing

- The main **objective of Engineering Drawing** are:
- To understand and apply the principles of engineering drawing as a graphic language used in industry to communicate technical information. This includes:
  - ✓ Visualization – Developing the ability to mentally manipulate and interpret visual information.
  - ✓ Graphics Theory – Understanding geometry and projection methods used to represent objects.
  - ✓ Standards – Applying the rules that ensure consistency in manufacturing and drawing practices.
  - ✓ Conventions – Applying commonly accepted methods in technical drawings.
  - ✓ Tools – Using instruments to create accurate technical drawings and models.
  - ✓ Applications – Exploring the practical uses of engineering drawings across various engineering fields.

1.3 Lettering

Text on engineering drawing is used :

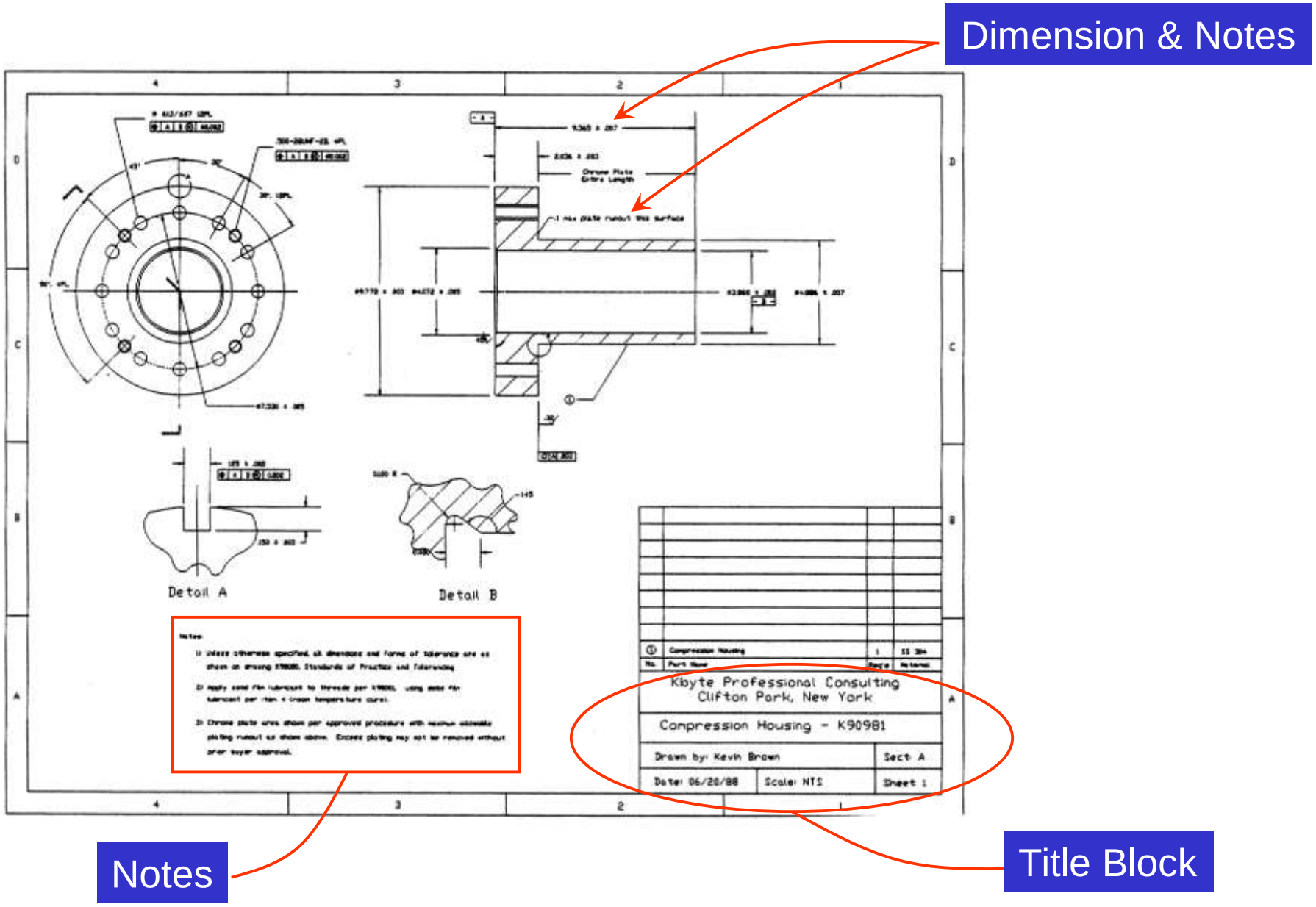
- ✓ To communicate nongraphic information.
- ✓ As a substitute for graphic information, in those instance where text can communicate the needed information more clearly and quickly.

Thus, it must be written with

Legibility - shape
- space between letters and words

Uniformity - size
- line thickness

Example *Placement of the text on drawing*



Lettering styles

ABCDEFGH
abcdefgh

Roman - All letters having elementary strokes " accented " or consisting of heavy and light lines, are classified as Roman

ABCDEFGH
abcdefgh

Text – this term includes all styles of old English. German text, Bradley text or others of various trade names – Text styles
Are too illegible for commercial purposes.

ABCDEFGH
abcdefgh

GOTHIC – All letters having the elementary Strokes of even width are classified as Gothic

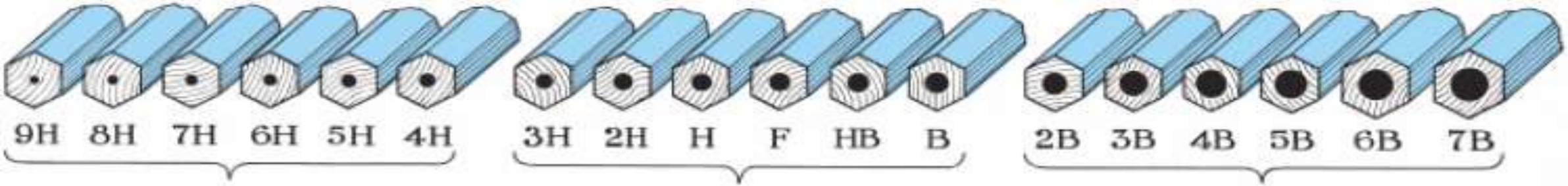
ABCDEFGH
abcdefgh

Italic – All slanting letters are classified as Italics. These may be further designated as Roman Italics, Gothic Italics or Text Italic.

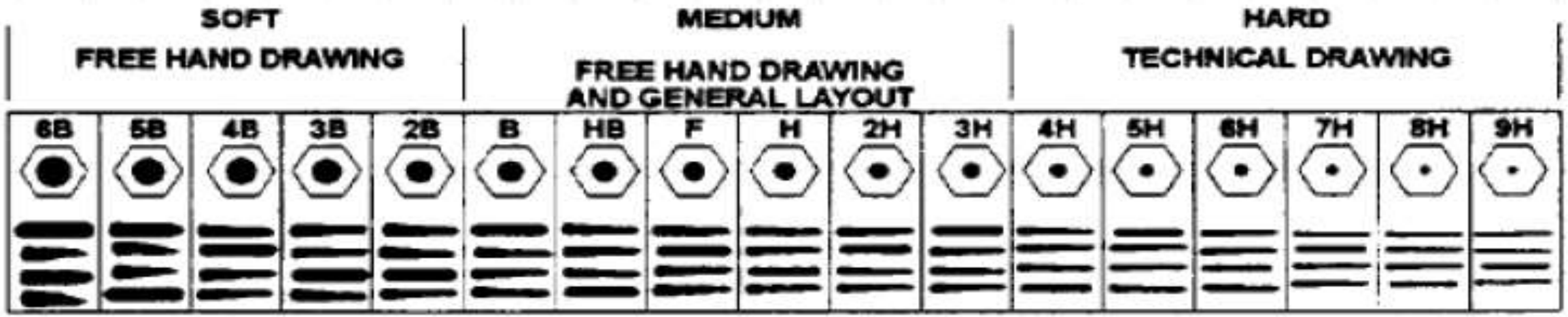
Free hand lettering

- Developing good lettering is a personal skill which takes practice. Letters must be formed properly.
- They must be open shaped very legible.
- Clear communication is essential to avoid mistakes and reduce waste.
- Lettering may be done using a drafting type pencil, lead holder or technical pen. Which ever tool is used, the letters must properly formed and very black

Pencil Lead Designation



- Note: Reverse order below



Pencil Techniques

- The best pencil for lettering on most surfaces are the H, F, and HB grades.
- Hold your pencil in the position shown. It should make approximately a 60° angle with the paper.

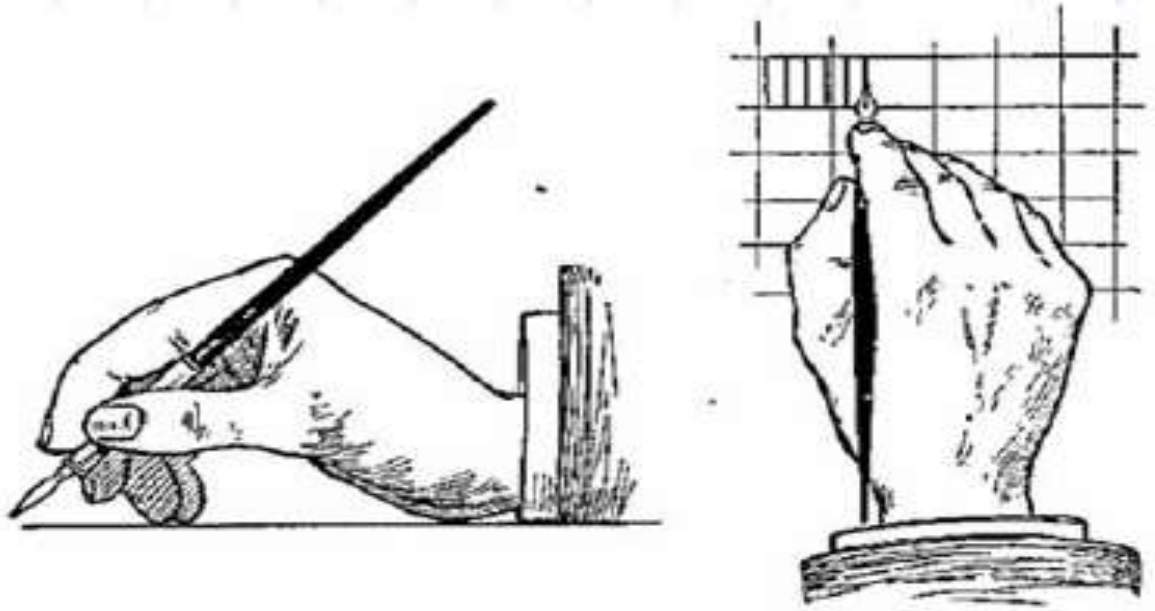


FIG. 105.—Position for lettering.

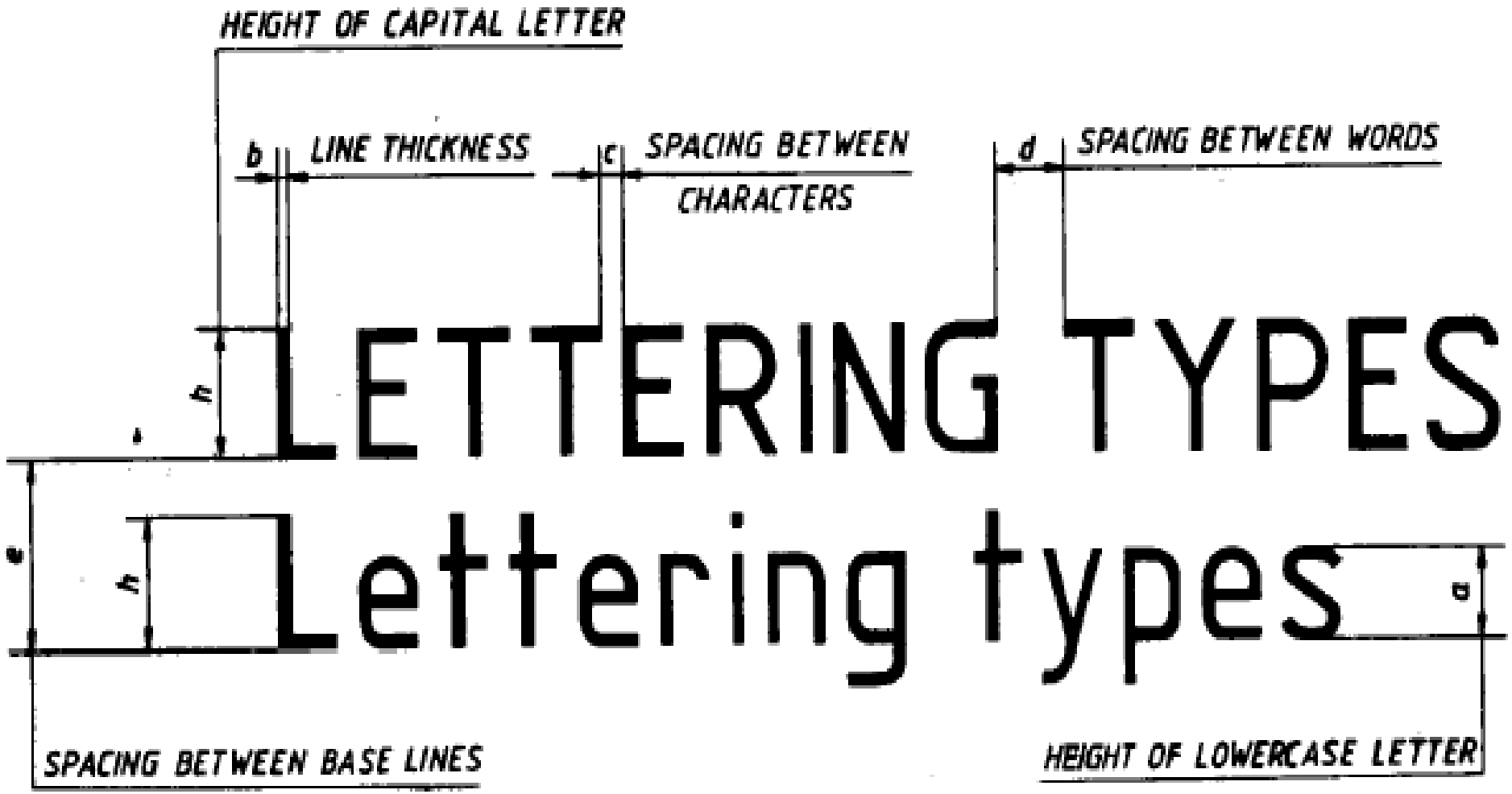
Standards

Standard Codes

Country	Code	Full name
USA	ANSI	American National Standard Institute
Japan	JIS	Japanese Industrial Standard
UK	BS	British Standard
Australia	AS	Australian Standard
Germany	DIN	Deutsches Institut für Normung
	ISO	International Standards Organization

Lettering types

- **Lettering A** – Height of the capital letter is divided into 14 equal parts
- **Lettering B** – Height of the capital letter is divided into 10 equal parts



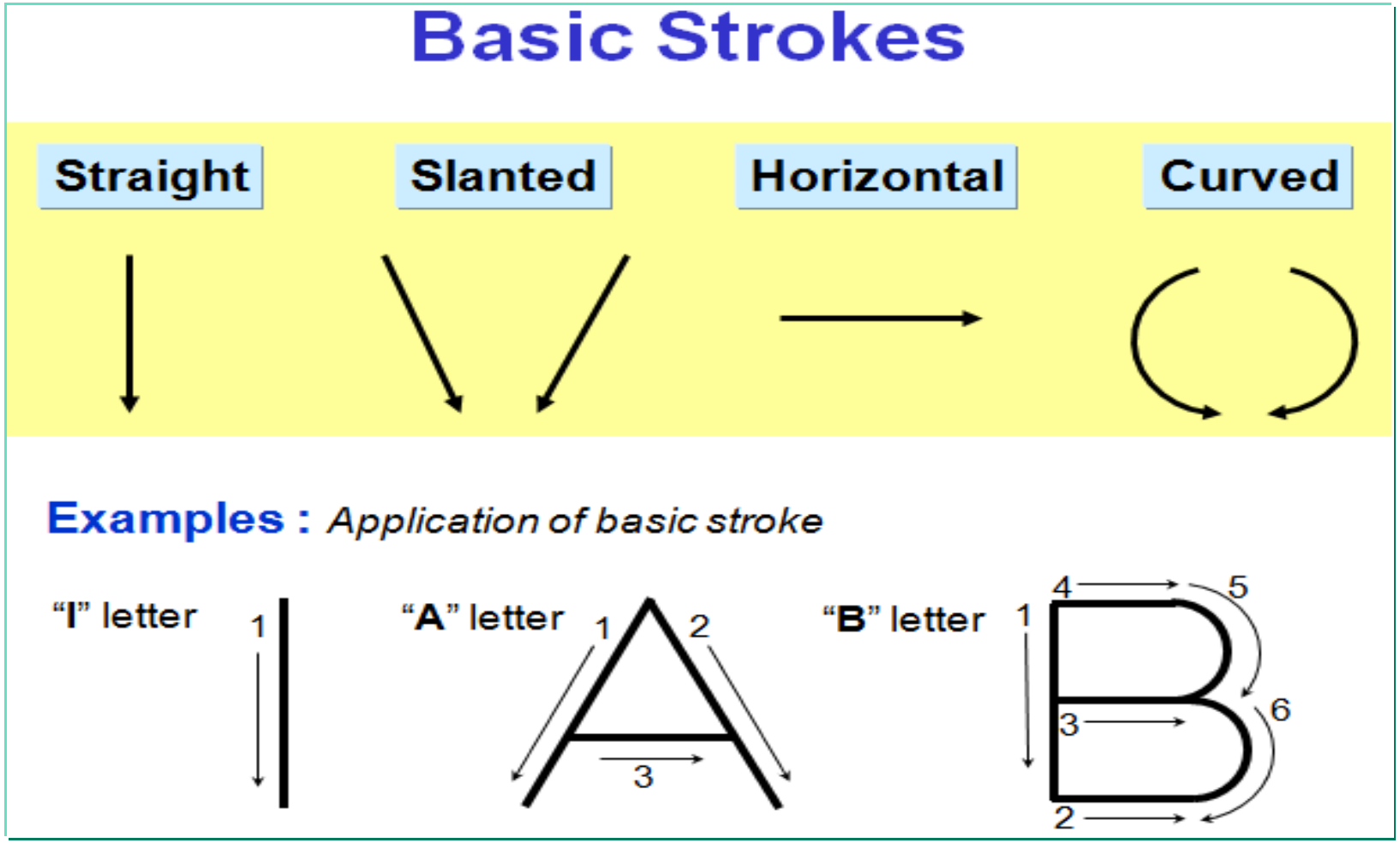
Specifications of A -Type Lettering

Specifications	Value	Size (mm)						
Capital letter height	h	2.5	3.5	5	7	10	14	20
Lowercase letter height	a = (5/7)h	-	2.5	3.5	5	7	10	14
Thickness of lines	b = (1/14)h	0.18	0.25	0.35	0.5	0.7	1	1.4
Spacing between characters	c = (1/7)h	0.35	0.5	0.7	1	1.4	2	2.8
Min. spacing b/n words	d = (3/7)h	1.05	1.5	2.1	3	4.2	6	8.4
Min. spacing b/n baselines	e = (10/7)h	3.5	5	7	10	14	20	28

Specifications of B -Type Lettering

Specifications	Value	Size (mm)						
Capital letter height	h	2.5	3.5	5	7	10	14	20
Lowercase letter height	a = (7/10)h	-	2.5	3.5	5	7	10	14
Thickness of lines	b = (1/10)h	0.25	0.35	0.5	0.7	1	1.4	2
Spacing between characters	c = (1/5)h	0.5	0.7	1	1.4	2	2.8	4
Min. spacing b/n words	d = (3/5)h	1.5	2.1	3	4.2	6	8.4	12
Min. spacing b/n baselines	e = (7/5)h	3.5	5	7	10	14	20	28

Single stroke lettering for use in engineering drawing – width of the stem of the letters and numerals will be uniformly thick equal to thickness of lines produced by the tip of the pencil.

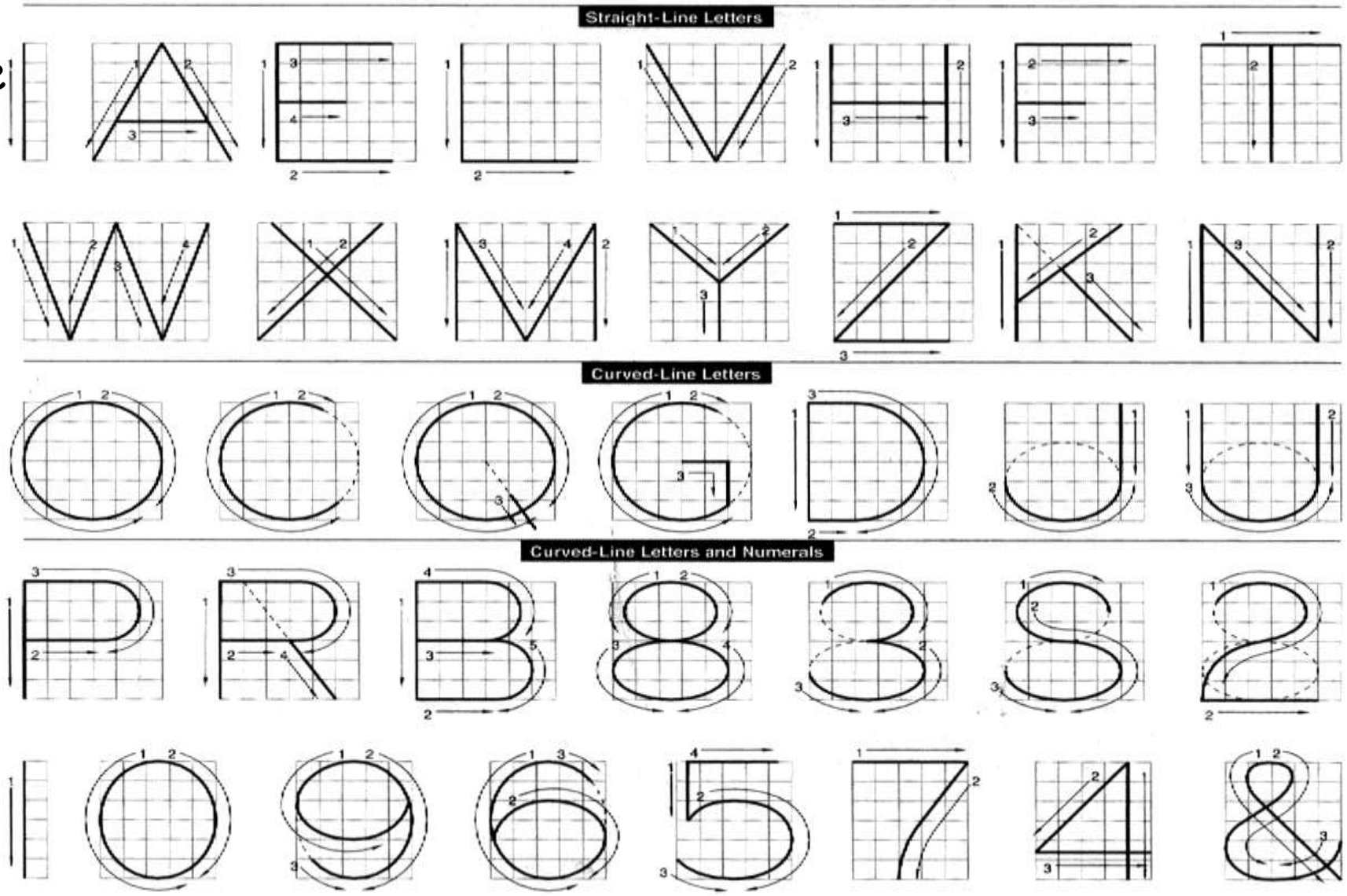


Upper-case letters & Numerals

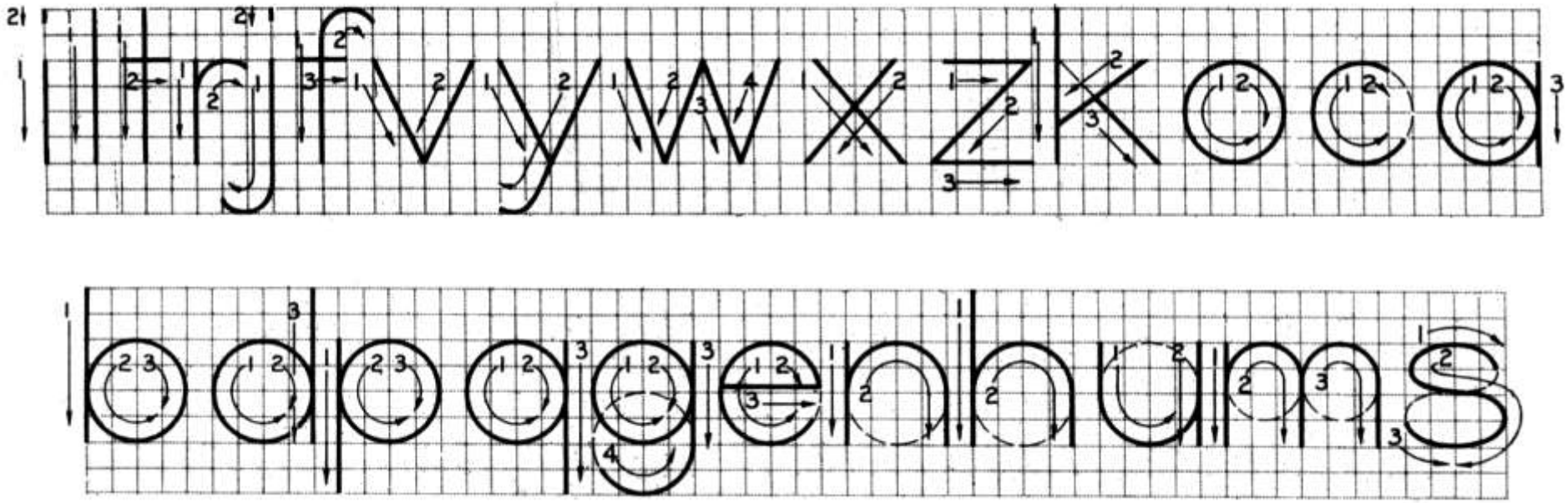
Straight line letters

Curved line letters

Curved line letters & Numerals



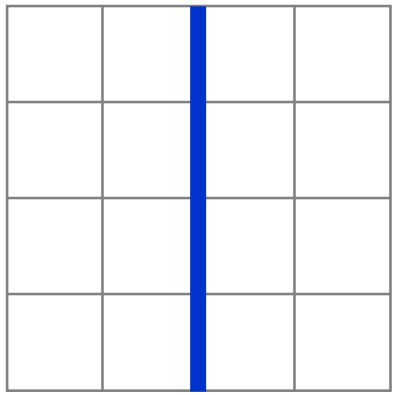
Lower-case letters



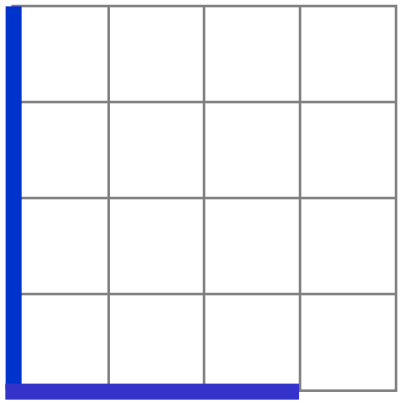
- The text' s body height is about 2/3 the height of a capital letter.

Stroke Sequence

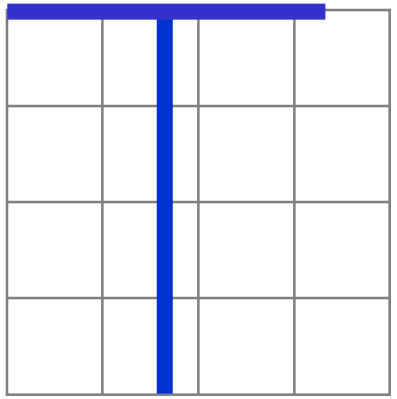
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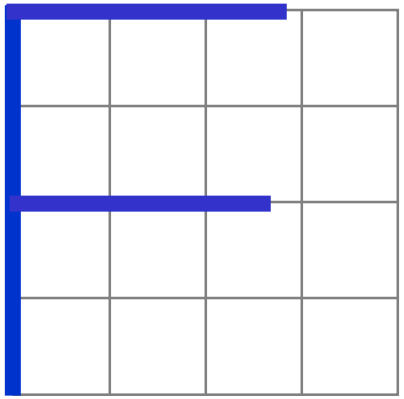
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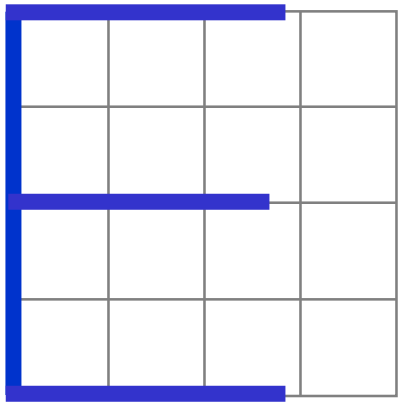
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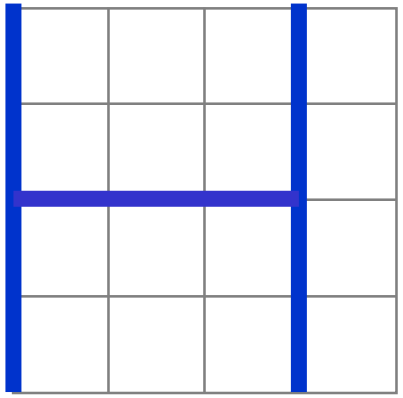
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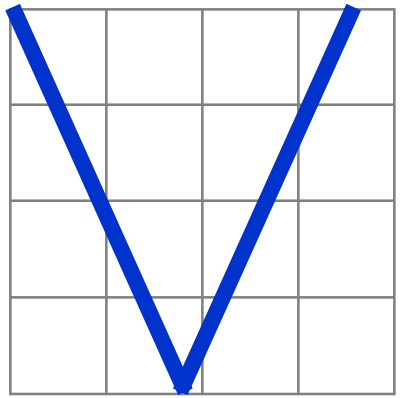


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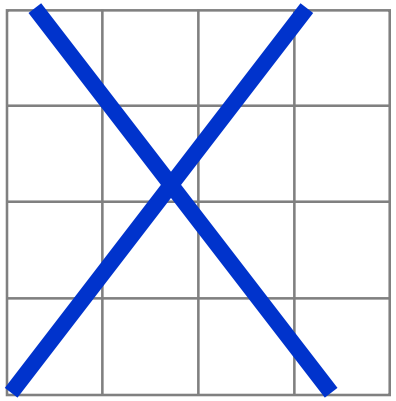


Stroke Sequence

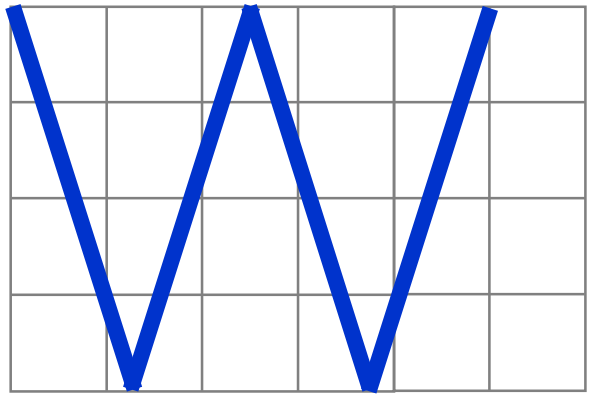
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X

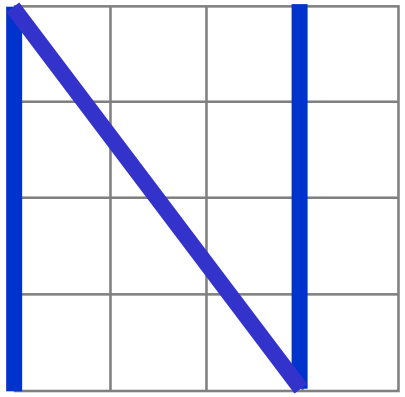


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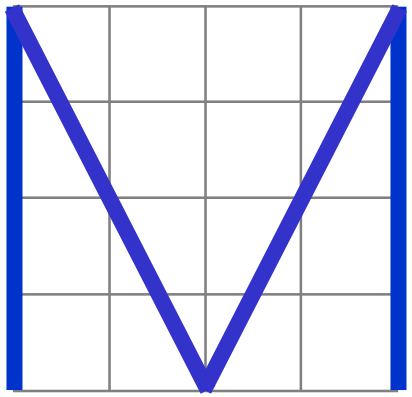


Stroke Sequence

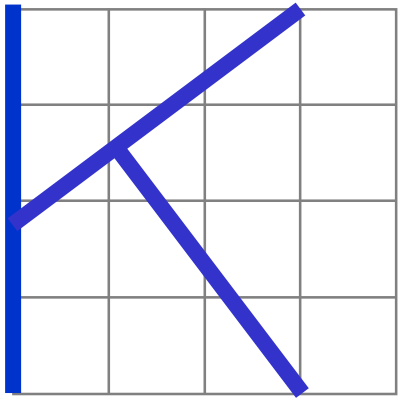
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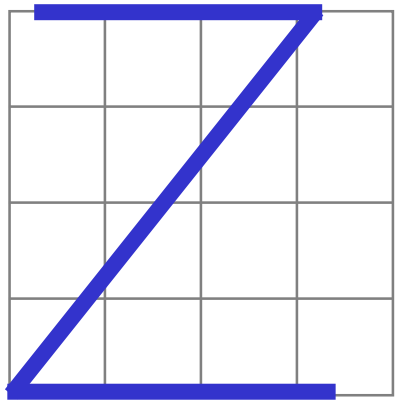
M



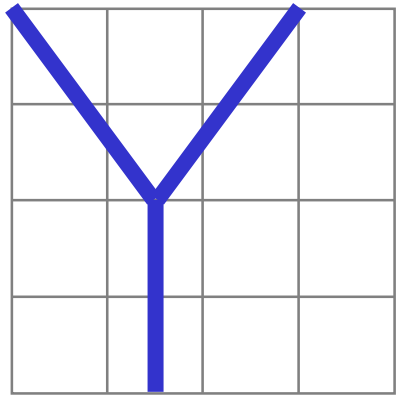
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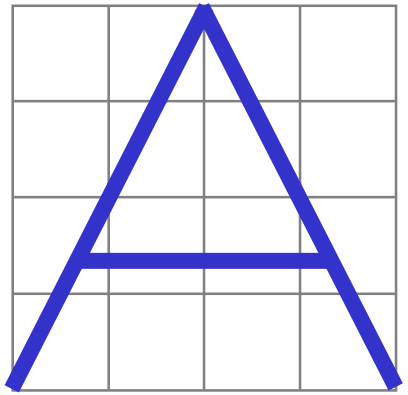
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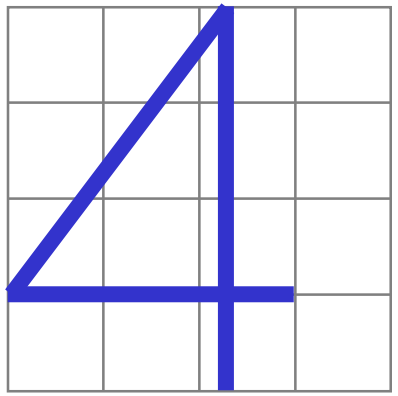
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A

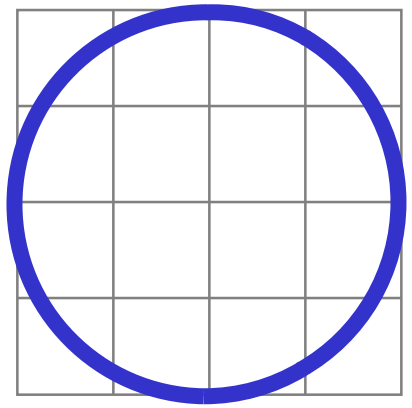


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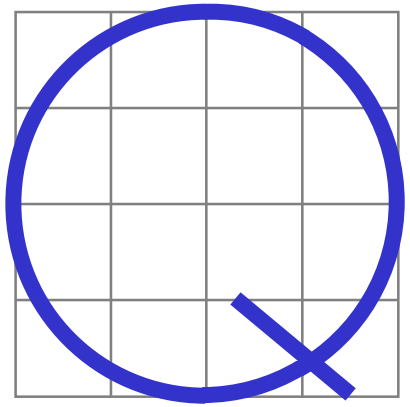


Stroke Sequence

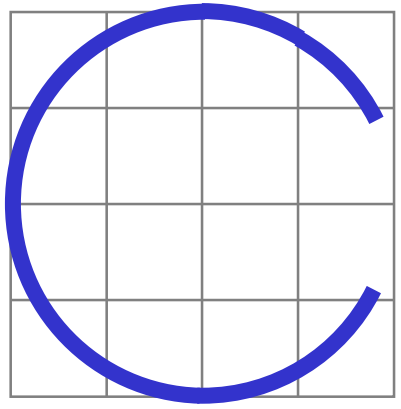
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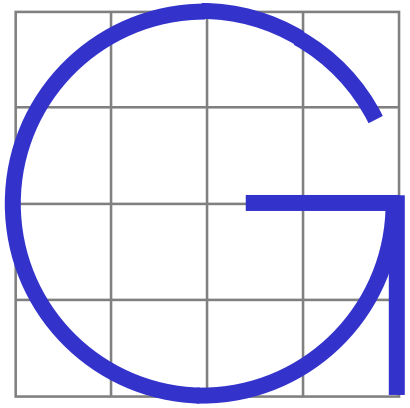
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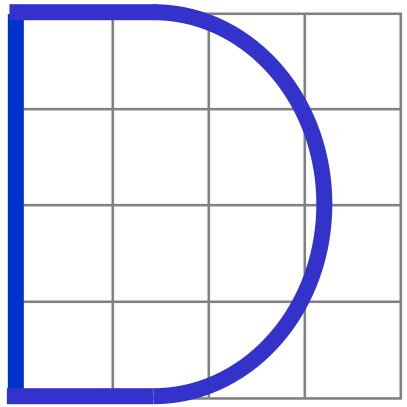


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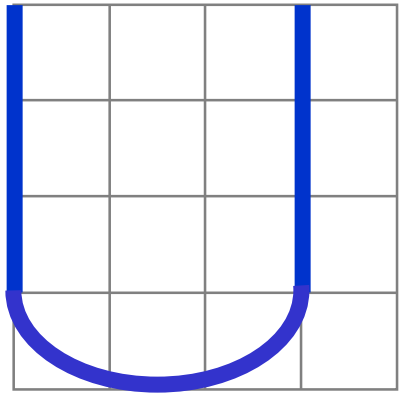


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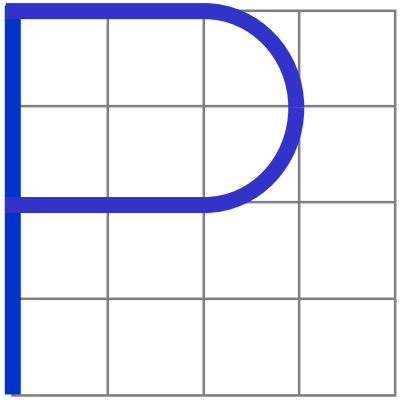
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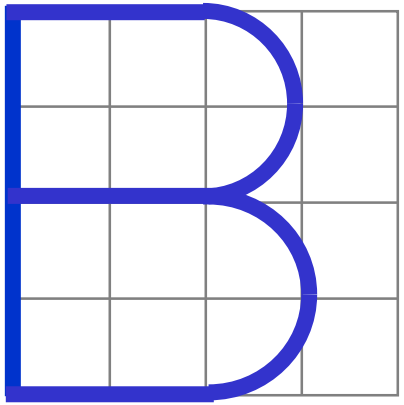
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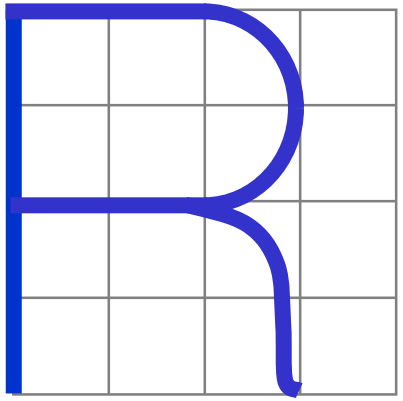
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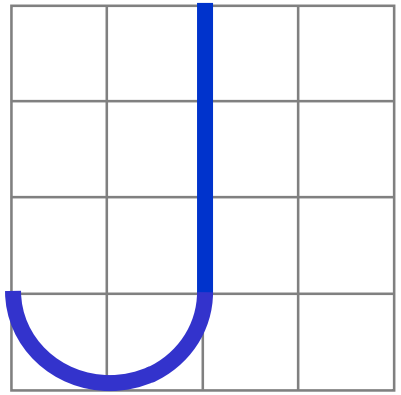
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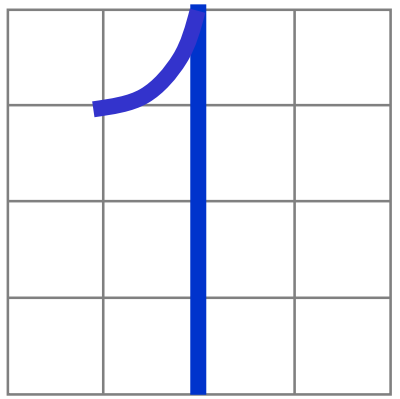
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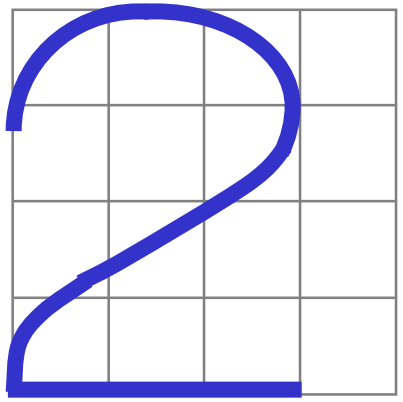
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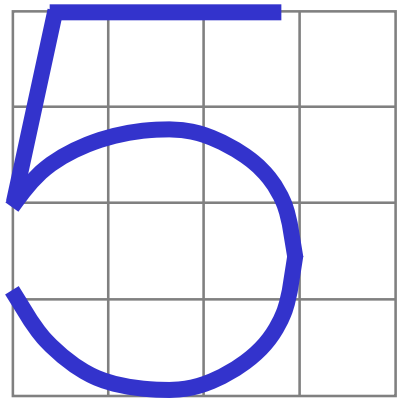


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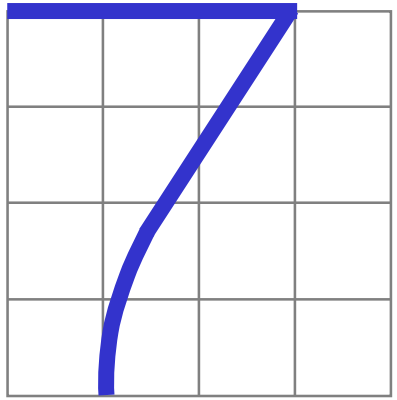


Stroke Sequence

5

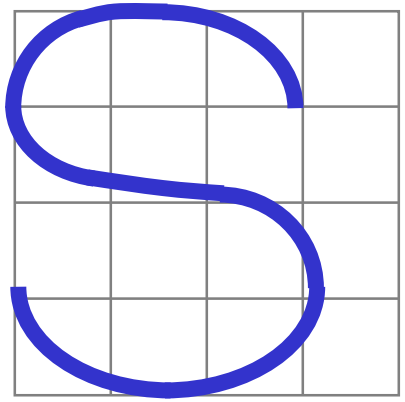


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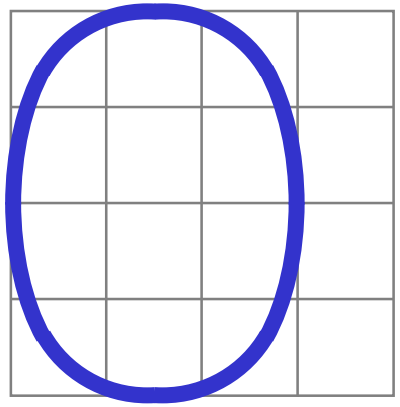


Stroke Sequence

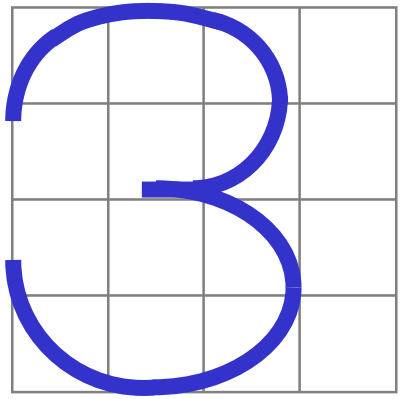
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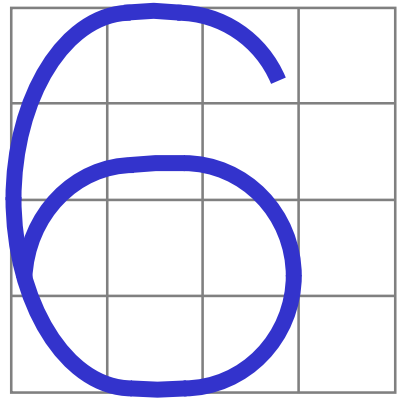
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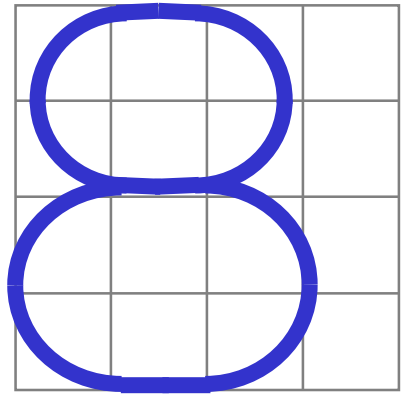
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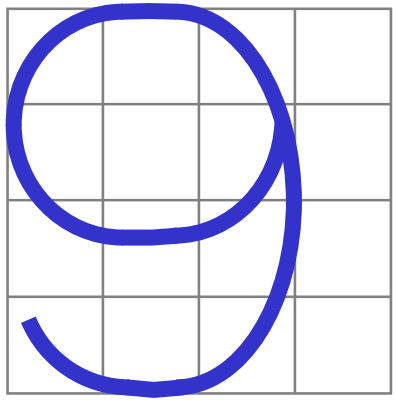
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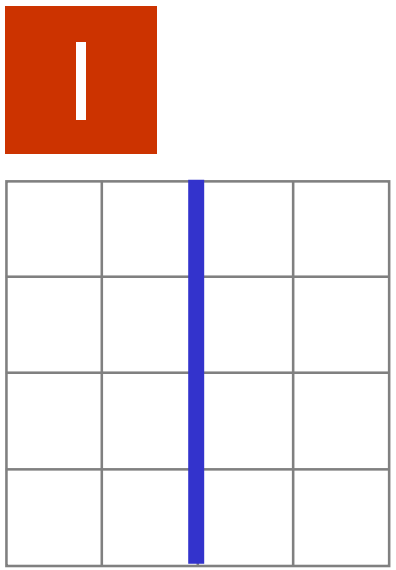
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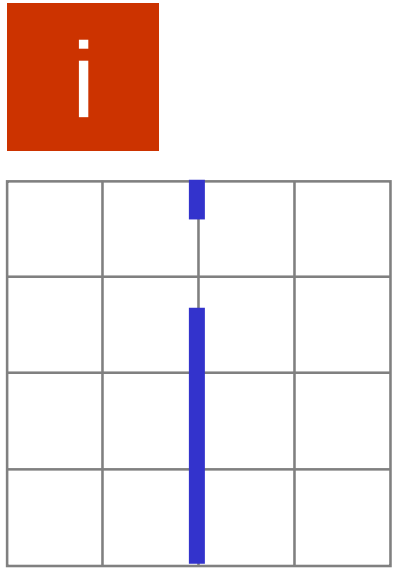


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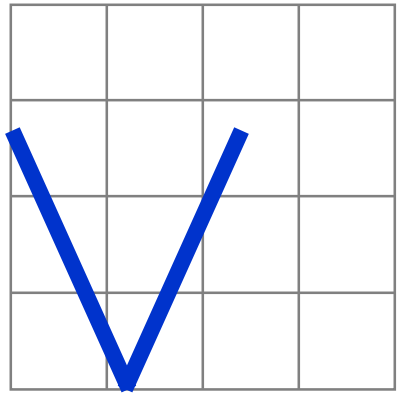
Stroke Sequence



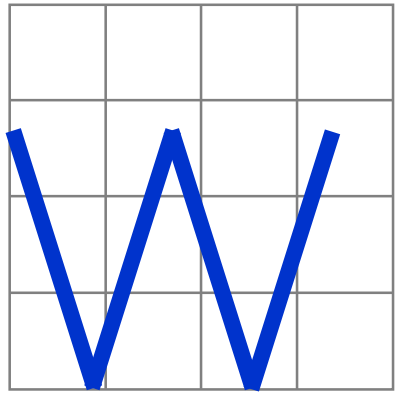


Stroke Sequence

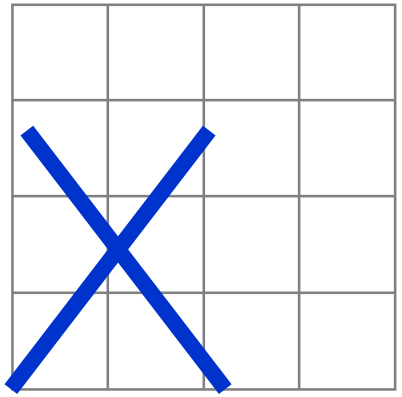
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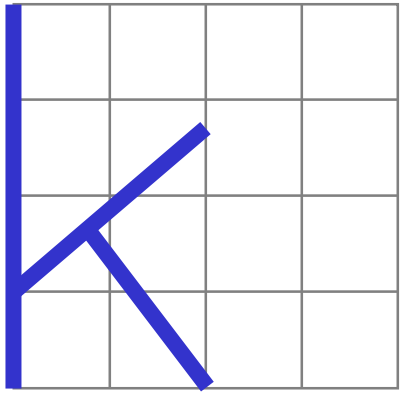
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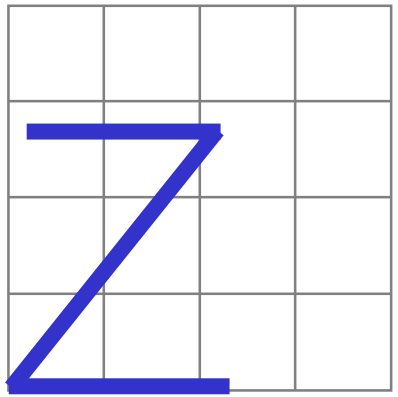
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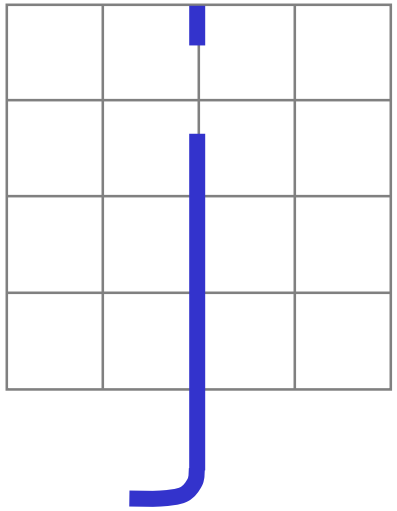


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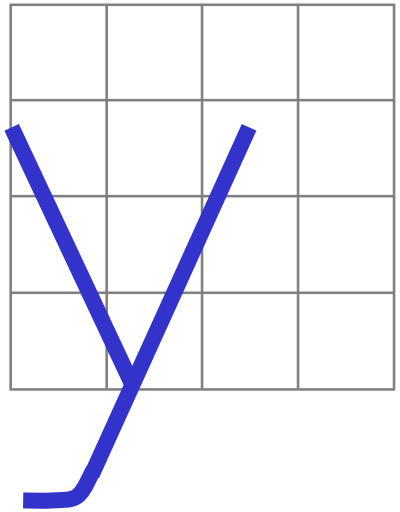


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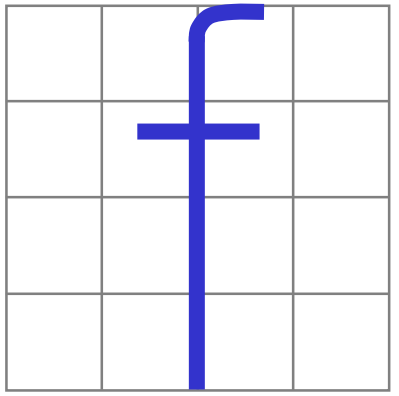
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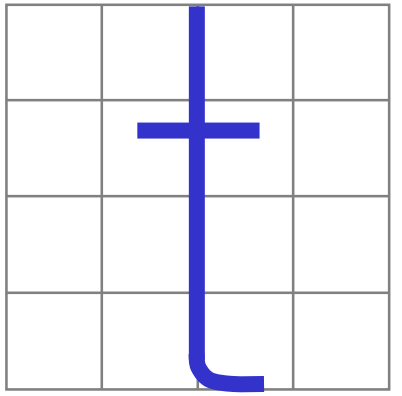
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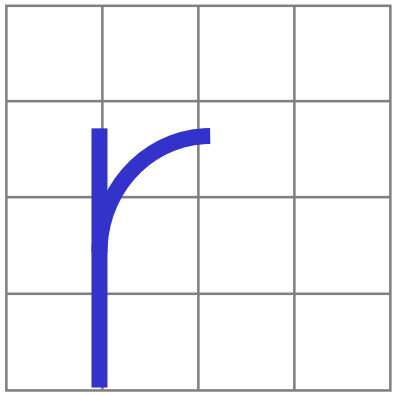
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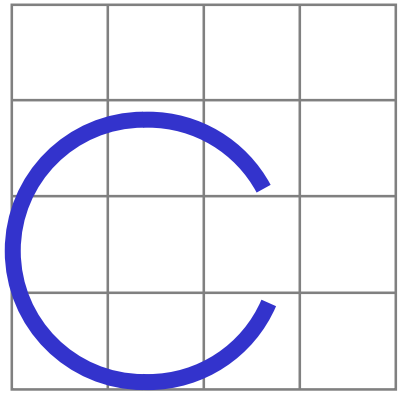


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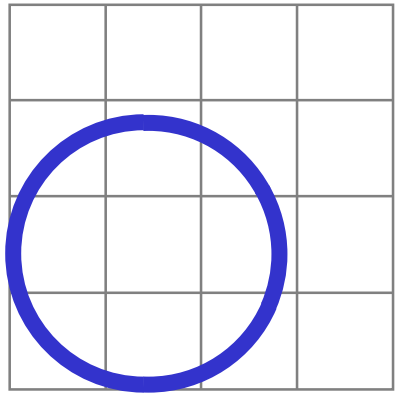


Stroke Sequence

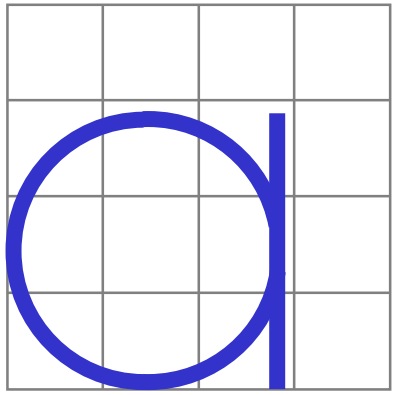
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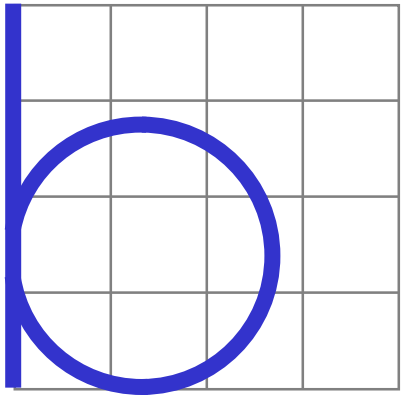
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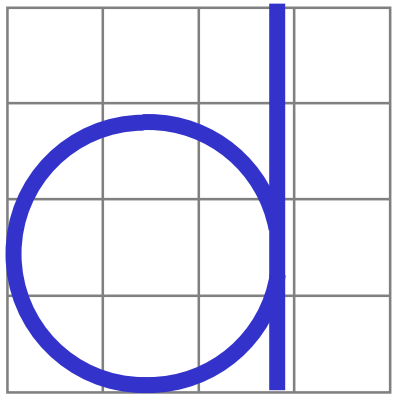
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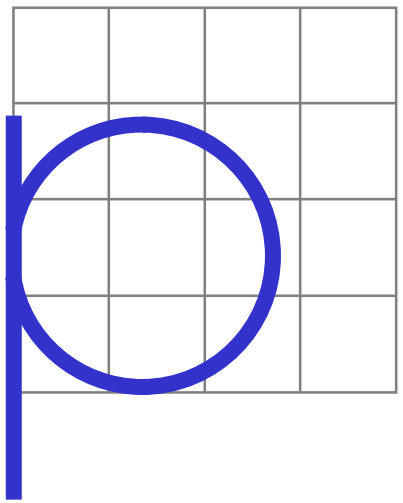
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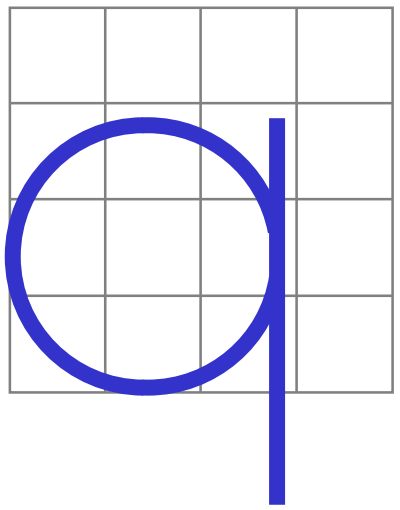
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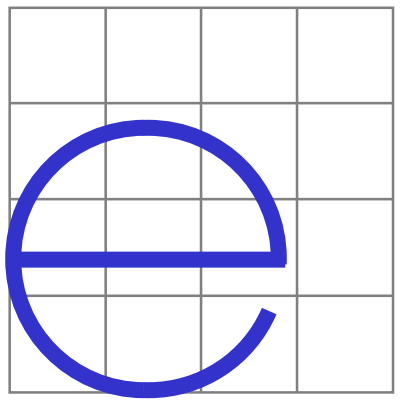
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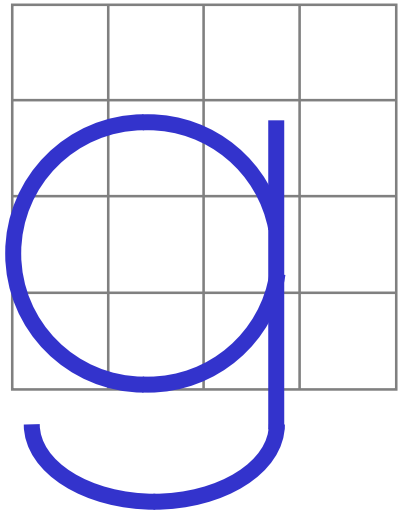


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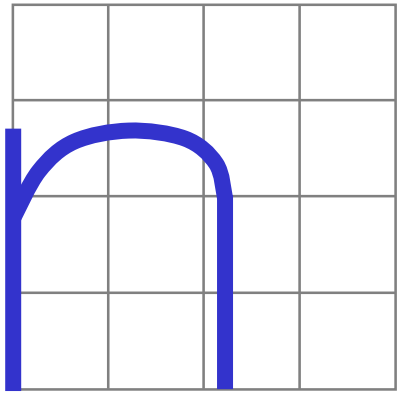


Stroke Sequence

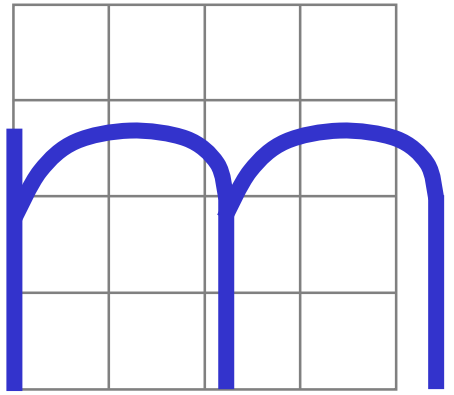
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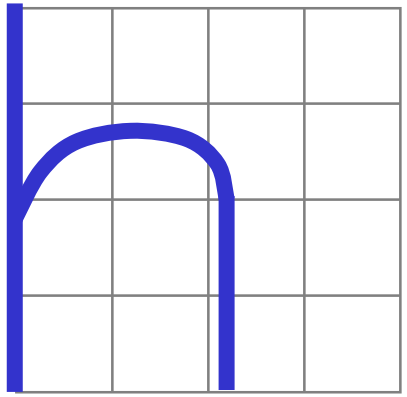
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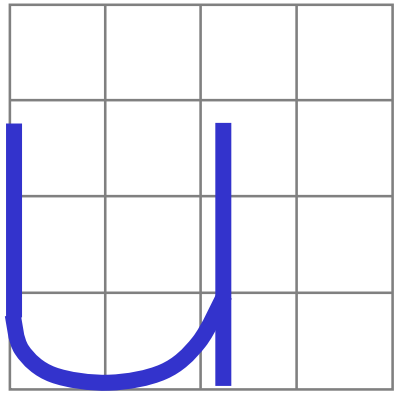
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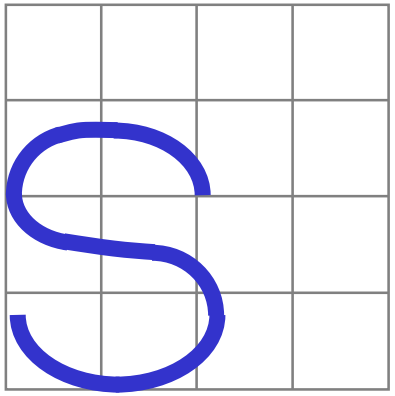
h



u



s



Example : *Good and Poor Lettering*



GOOD

Not uniform in style.

Not uniform in height.

Not uniformly vertical or inclined.

Not uniform in thickness of stroke.

Area between letters not uniform.

Area between words not uniform.

1.4 Drawing Instruments

- Drawings can be created quickly and precisely using drawing tools.
- The following are the minimal drawing equipment and other drawing materials that every student must have:

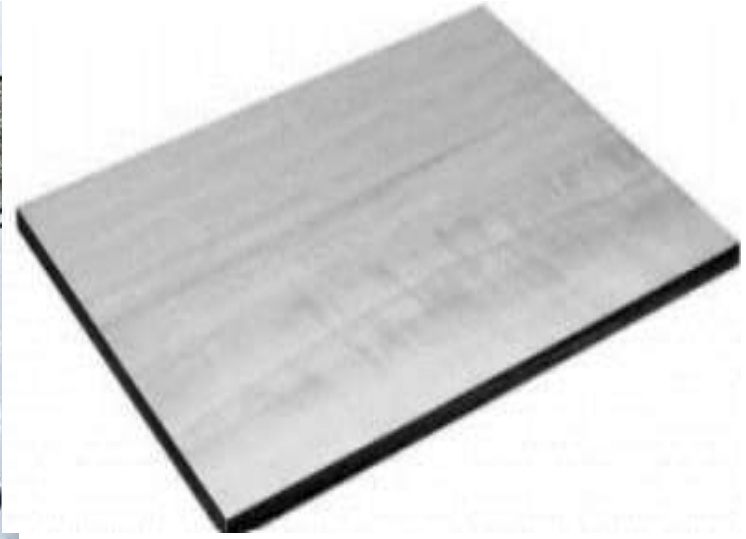
1. Drawing board
2. T-square
3. Set-squares–45° and 30°-60°
4. Drawing instrument box, containing: Compass, Divider and Inking pen
5. Scales
6. Protractor
7. French curves
8. Drawing papers
9. Drawing pencil
10. Eraser (Rubber)
11. Adhesive tapes
12. Duster
13. Circle templates
14. Sharpener

Drawing Instruments…

- **Drawing board:** This is a wooden plate form on which the drawing paper is placed.

SIZES OF DRAWING BOARDS

Designation	Size (mm)
B0	1000 × 1500
B1	700 × 1000
B2	500 × 700
B3	350 × 500





- **T-square:** It is used for drawing horizontal lines and it also support set-squares during drawing.



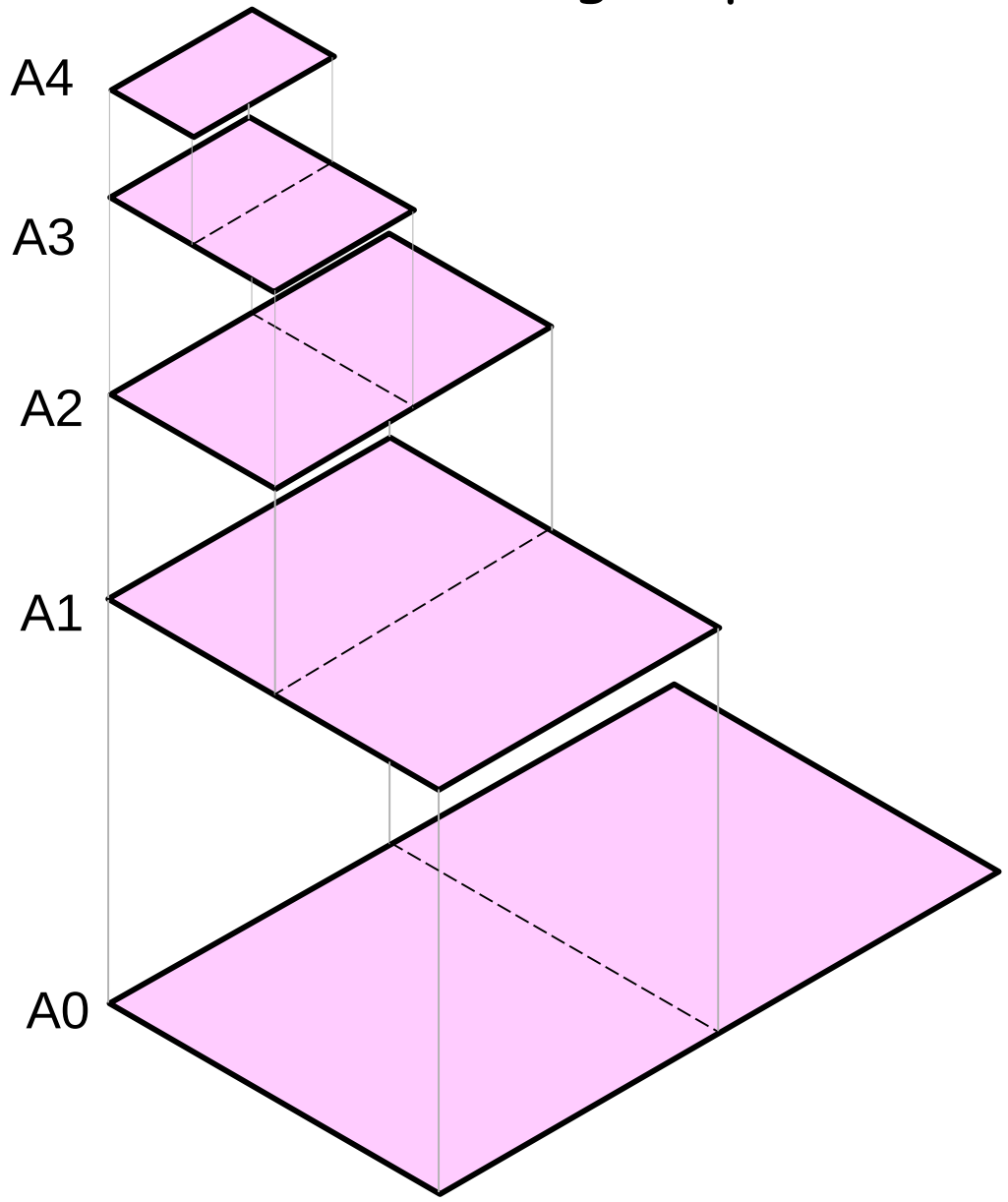
Drawing Instruments…

- Drawing paper: It is the paper on which the drawing is produce. Trimmed paper suitable sizes are A0 ~ A4.

Standard sheet size (**ISO**)

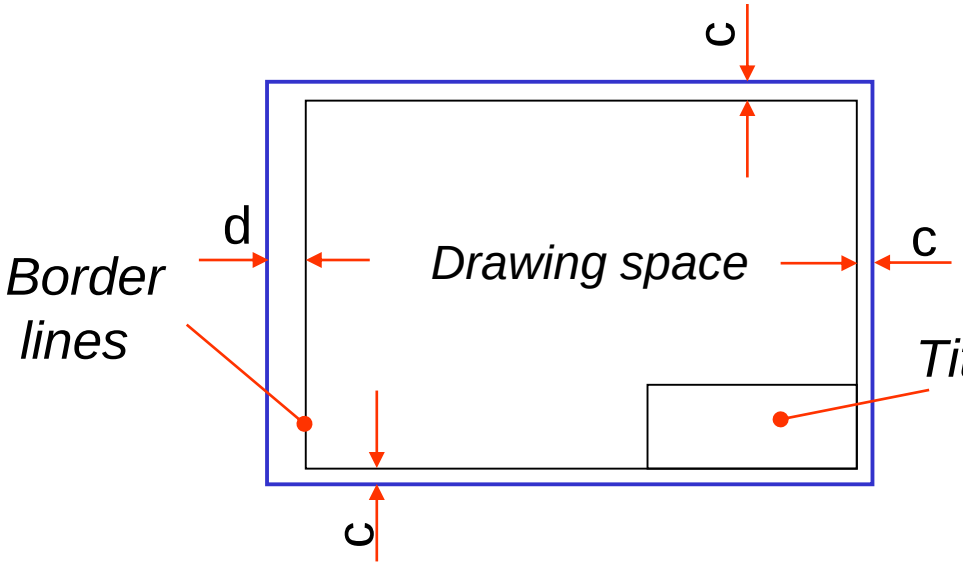
A4	210 x 297
A3	297 x 420
A2	420 x 594
A1	594 x 841
A0	841 x 1189

(Dimensions in millimeters)

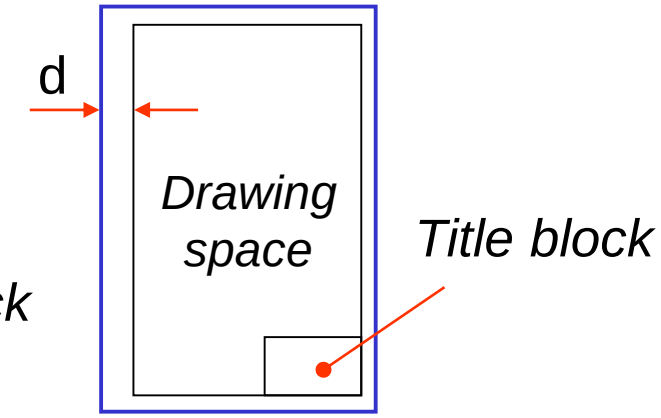


Orientation of drawing sheet

1. Type X (A0~A4)



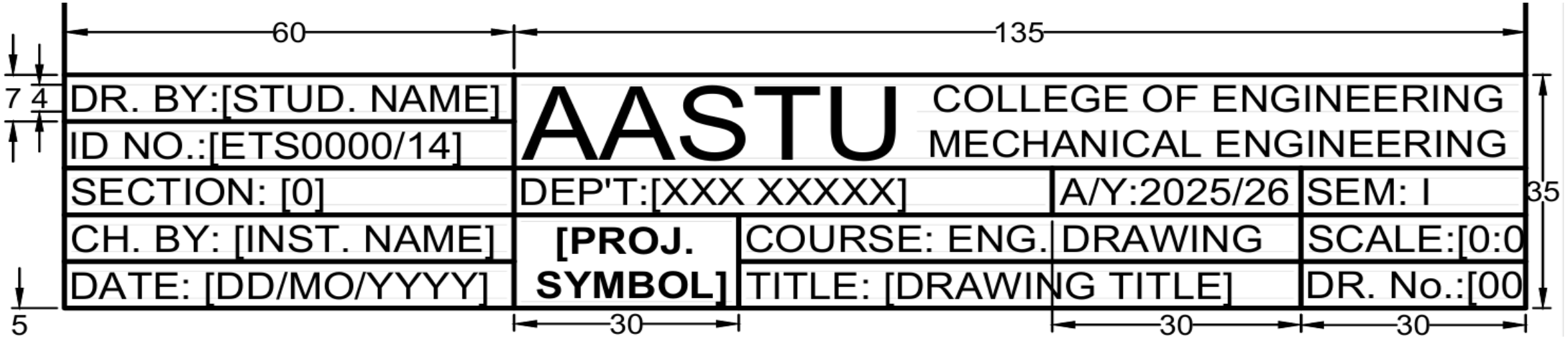
2. Type Y (A4 only)



Sheet size	c (min)	d (min)
A4	10	25
A3	10	25
A2	10	25
A1	20	25
A0	20	25

- Any engineering drawing contains a title block. The title block can have a maximum length of 170 mm providing the following information:
    1. Title of the drawing
    2. Drawing number
    3. Scale
    4. Symbol denoting the method of projection
    5. Name of the firm
    6. Initials of staff who have designed and checked
    7. Date

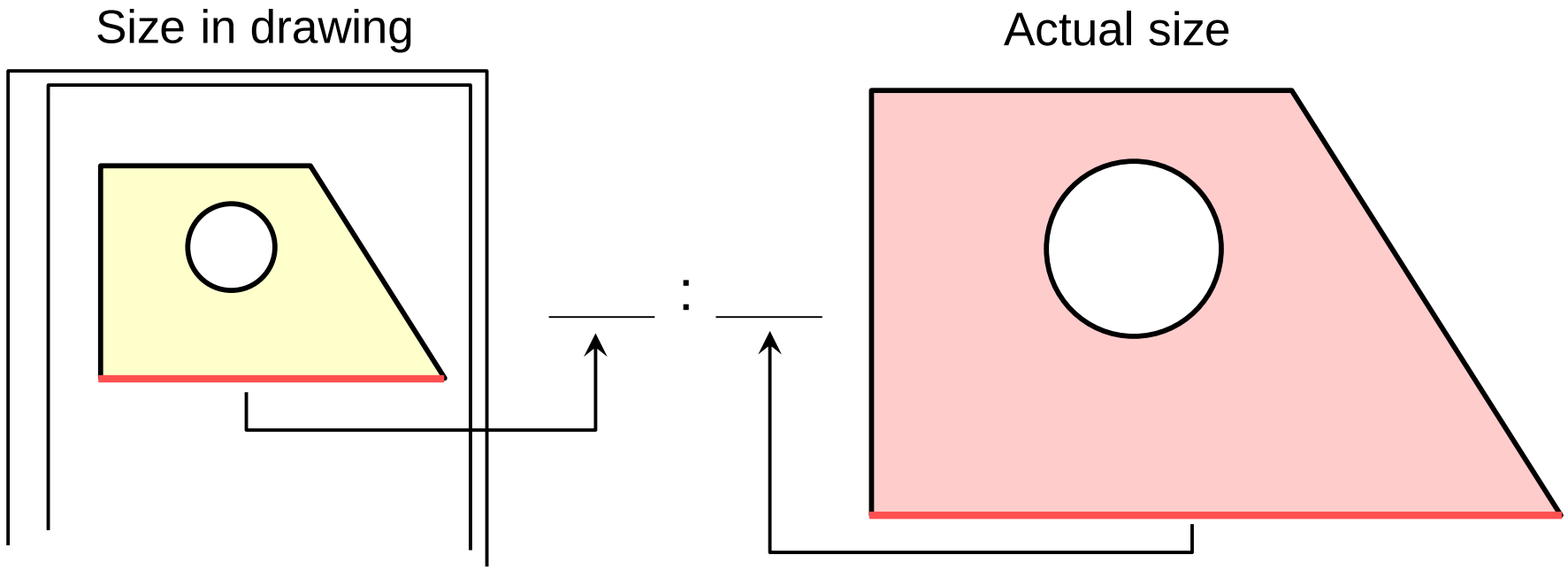
- **A simplified title block used for engineering Drawing**



Drawing Instruments…

Length, size

- **Scale** is the ratio of the linear dimension of an element of an object shown in the drawing to the real linear dimension of the same element of the object.



Drawing Instruments…

- Designation of a scale consists of the word "SCALE" followed by the indication of its ratio, as follow:

  SCALE 1:1 for full size

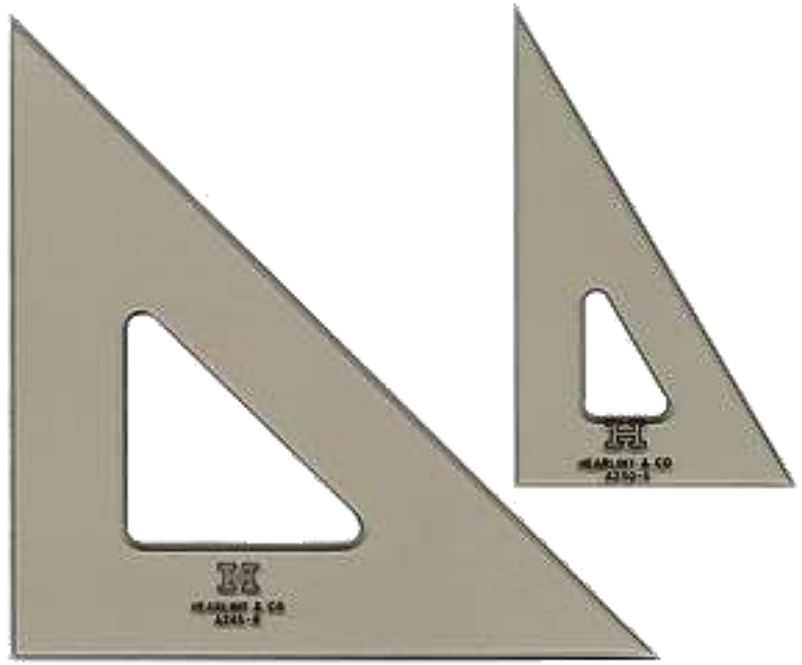
  SCALE **X**:1 for ***enlargement*** scales (**X** > 1)

  SCALE 1:**X** for ***reduction*** scales (**X** > 1)

- Dimension numbers shown in the drawing are correspond to "true size" of the object and they are independent of the scale used in creating that drawing.

Drawing Instruments…

- Set-squares: They are used for drawing vertical or diagonal lines. Two types of set-squares are (i) 30- and 60-degree set-square (ii) 45-degree set-square.



- Compass: It is used for drawing circles and arcs.
- Divider: It is used for transferring measurement and for setting distances.



Drawing Instruments…

- Drawing pencil: Two types of pencils are: hard pencils (H grade of pencils and soft pencil) and soft pencils (B grade of pencils). Three grades of pencils are used in engineering drawing. These are H, 2H and HB pencils.
  - ✓ H pencil is used for lettering and arrow heads.
  - ✓ 2H pencil is used for construction lines.
  - ✓ HB pencil is used for free hand sketching.





- Eraser: It is used for cleaning pencil. It is made of rubber.

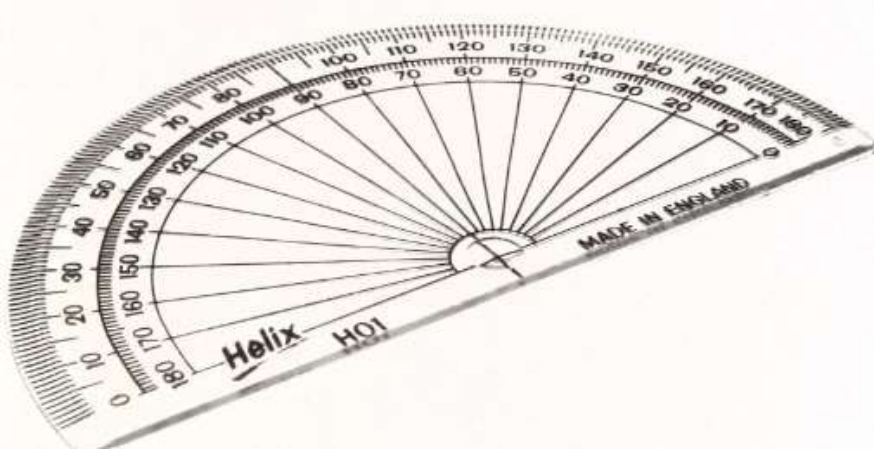


Drawing Instruments…

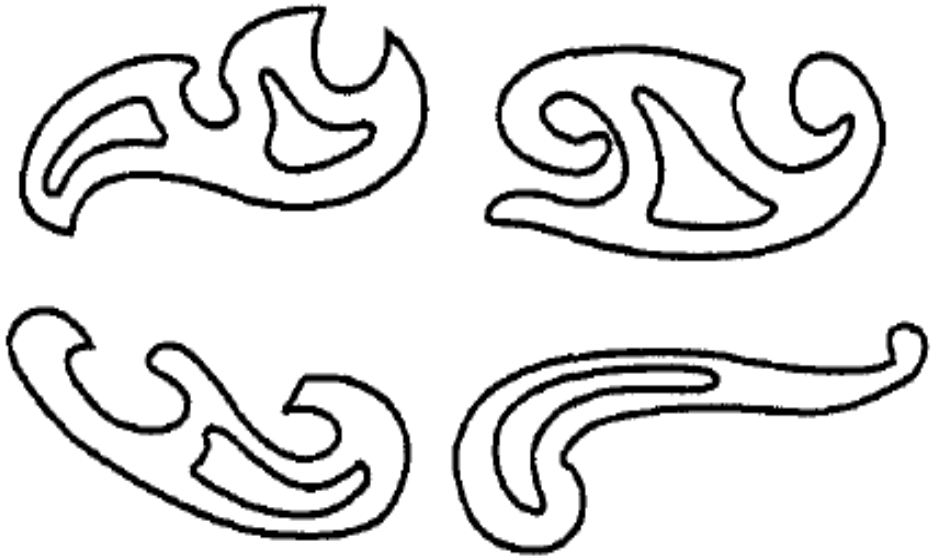
- Sharpener: It is used for sharpening pencils.



- Protractor: It is semi-circular in shape. It used for measuring angles.

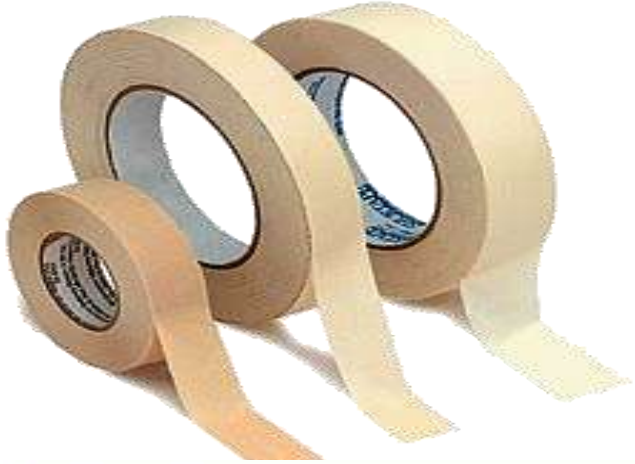


- French curve: It is used for drawing curves.

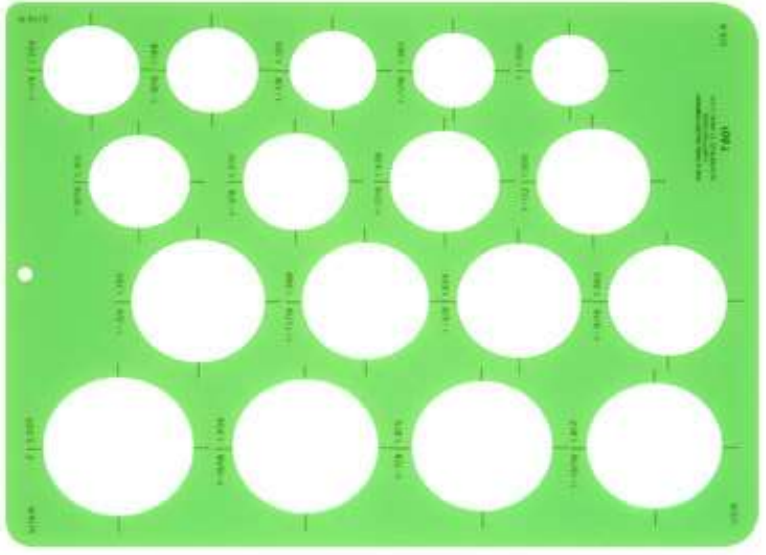


Drawing Instruments…

- Adhesive tape: Is used to fix the drawing paper on the drawing board.



- Circle templates: These are aids used for drawing small features such as circles, arcs and symbols.



- Duster: It should preferably be of towel cloth of convenient size used to clean all the instruments and materials.



1.5 Types of Lines

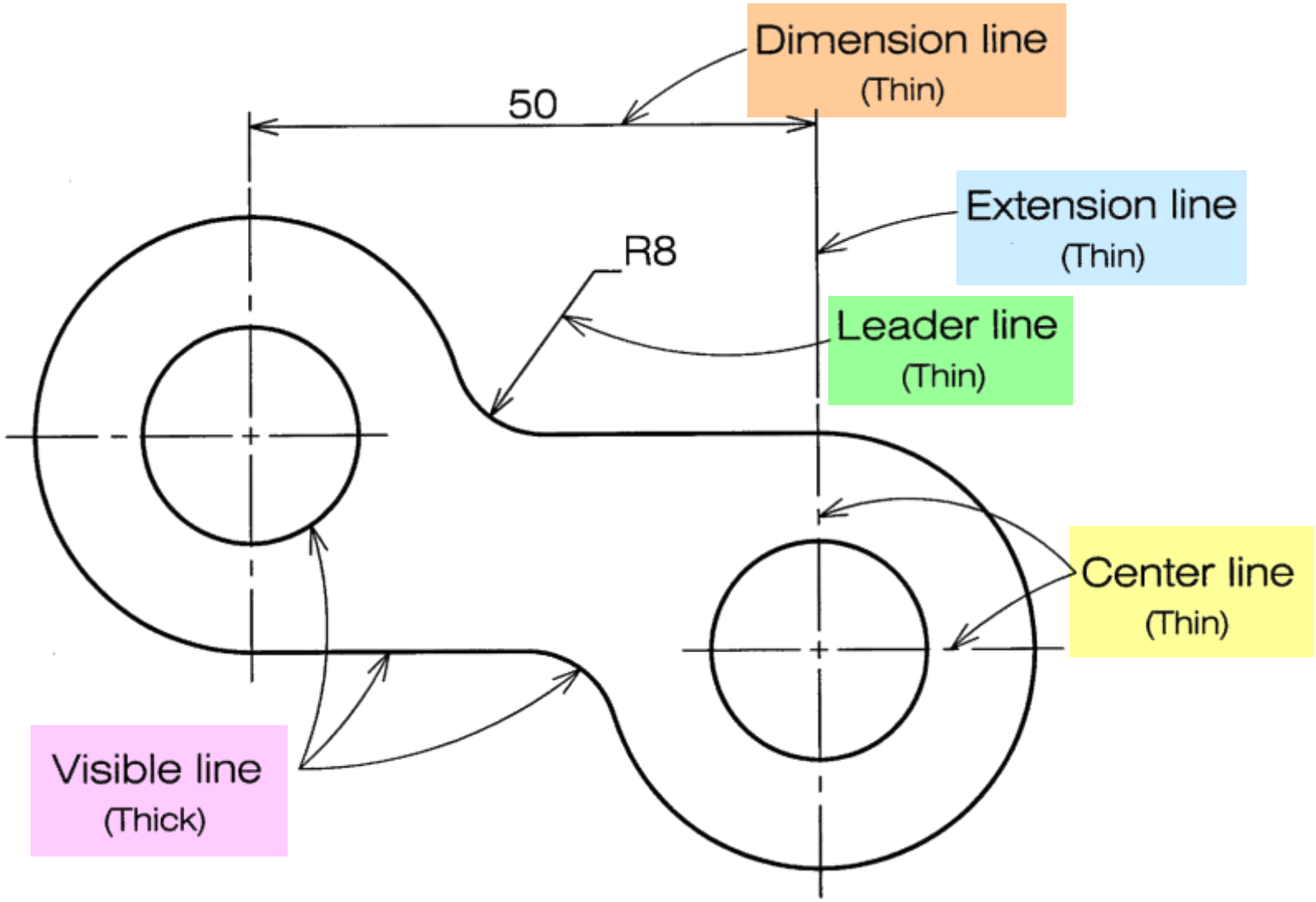
- Engineering drawings are prepared with the help of symbolic lines. The selection of thickness of the line depends on the type and size of drawing.
- The thickness of the line should be opted from one of the following:
  - ✓ 0.13 mm; 0.18 mm; 0.25 mm; 0.35 mm; 0.5 mm; 0.7 mm; 1 mm; 1.4 mm; 2 mm.

- ***Visible lines*** represent features that can be seen in the current view.
- ***Hidden lines*** represent features that can not be seen in the current view.
- ***Center line*** represents symmetry, path of motion, centers of circles, axis of symmetrical parts.
- ***Dimension and Extension lines*** indicate the sizes and location of features on a drawing.

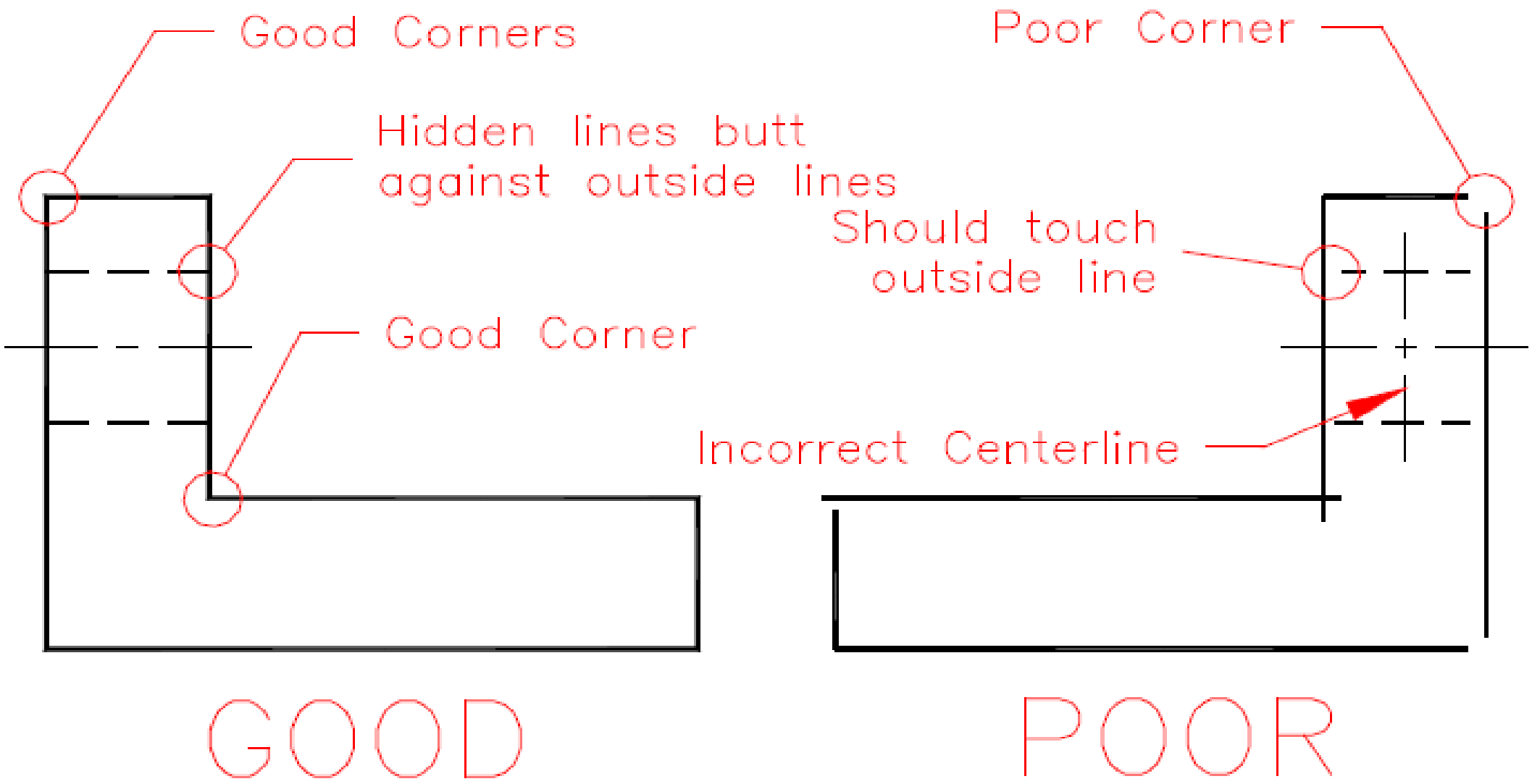
Common Drawing Elements and Recommended Practices

Drawing element	Recommended practice	Line style / illustration	Line thickness
1. Visible outlines	• Use continuous, prominent lines.		0.50 mm (thick line)
2. Construction lines	• Use light, thin lines where appropriate.		0.13 mm (thin line)
3. Hidden edges	• Use the prescribed dashed-line pattern.		0.25 mm (thin line)
4. Centerlines	• Use the prescribed chain-line pattern.		0.25 mm (thin line)
5. Dimension lines	• Use thin lines with suitable arrowheads or other standard terminations.		0.25 mm (thin line)
6. Extension lines	• Use thin lines to indicate dimension limits.		0.25 mm (thin line)
7. Cutting-plane lines	• Follow the applicable standard for line pattern, thickness, arrows, and labels.		0.35 mm (medium line)

Example : *Line conventions in engineering drawing*



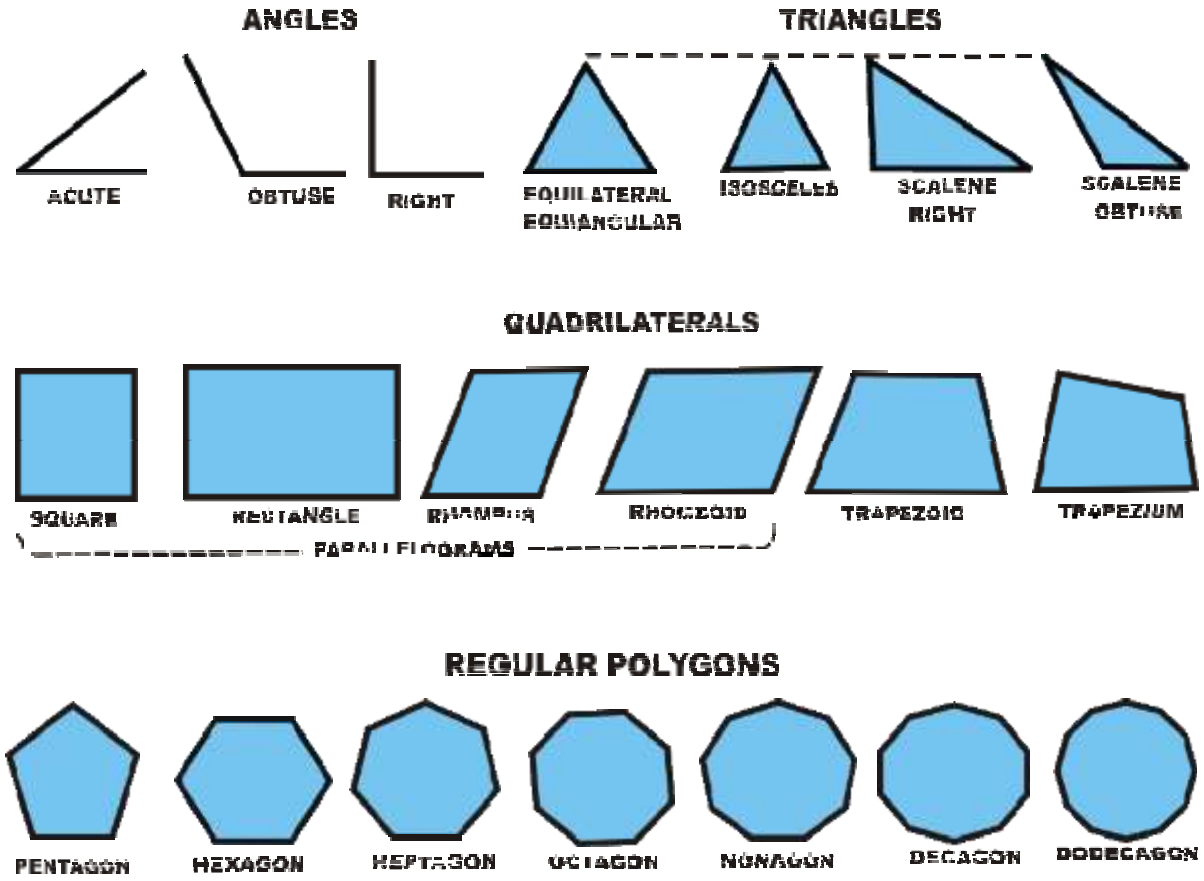
Good practice

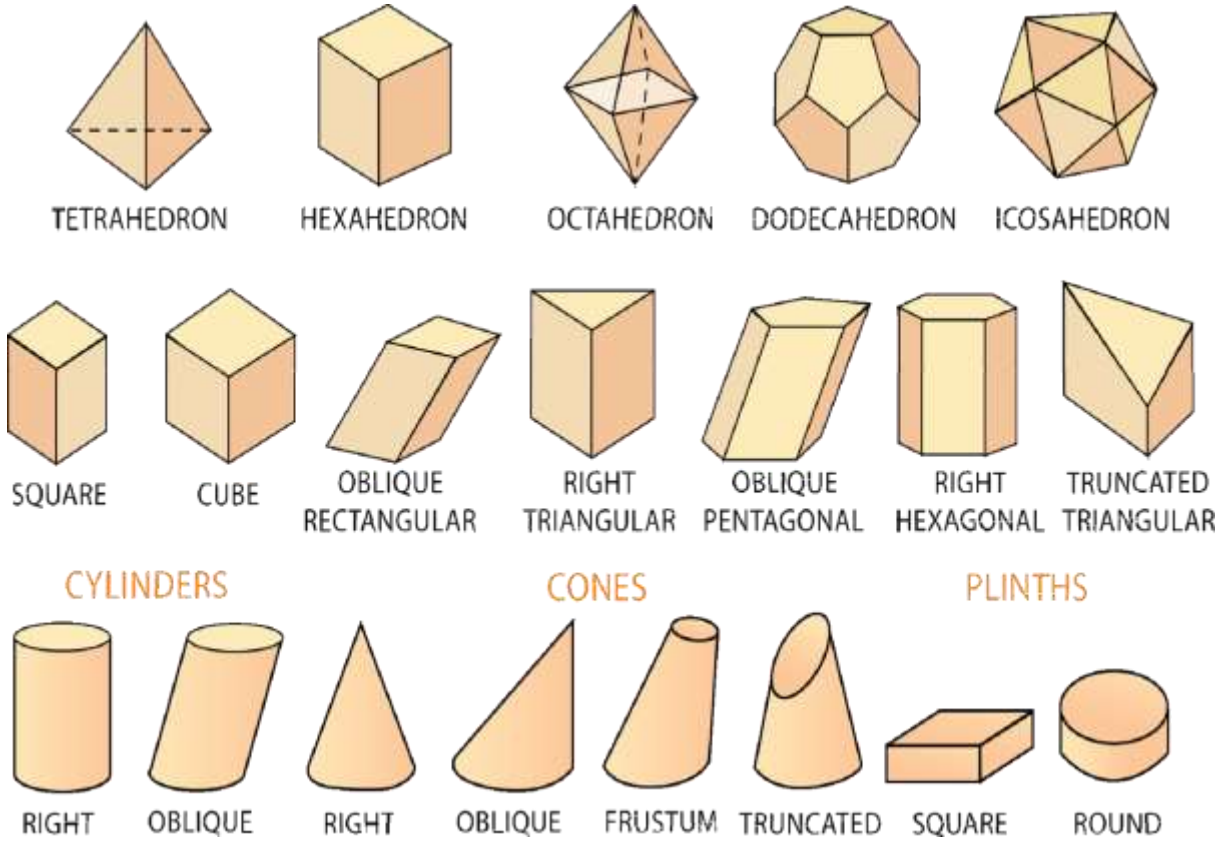


1.6 Geometric Construction

- The geometric construction forms the basis for all technical drawings.
- Geometric construction is the process of creating geometric figures accurately using drawing instruments and established geometric principles.
- Geometric construction involves creating points, lines, angles, circles, arcs, polygons, and curves according to specified dimensions or geometric relationships.
- Geometric construction is an essential foundation of engineering drawing because it enables engineers, designers, and technicians to represent shapes and relationships accurately.

- Example of geometric constructions:



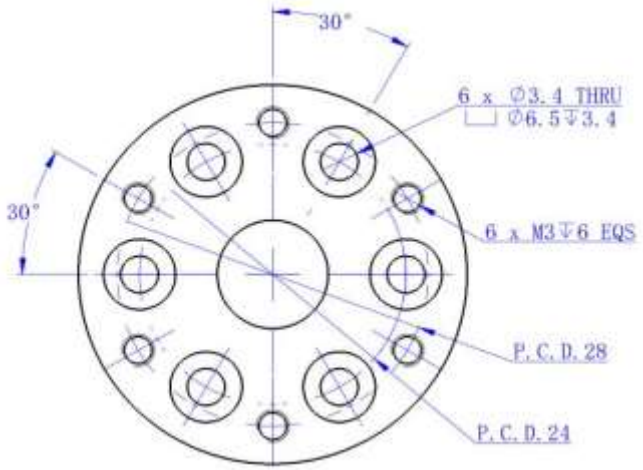


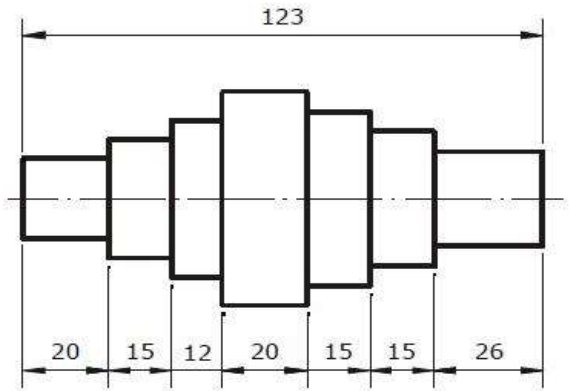
Applications in Different Engineering Fields

- Geometric construction is applied in mechanical, manufacturing, civil, and architectural engineering. Although the applications differ, the underlying principles are similar.

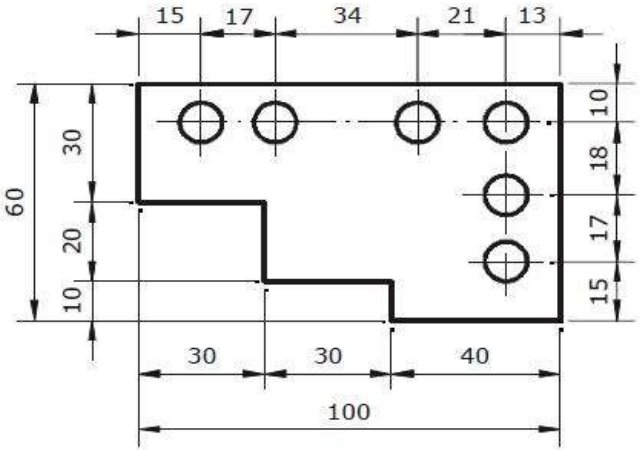
1. Mechanical engineering

- Common applications include:
  - ✓ Constructing shafts, gears, pulleys, flanges, and machine components.
  - ✓ Locating the centers of holes, circular features, and cylindrical parts.
  - ✓ Constructing fillets, rounded corners, and tangent connections.
  - ✓ Developing cam profiles and other mechanical curves.
  - ✓ Preparing orthographic views and component layouts.





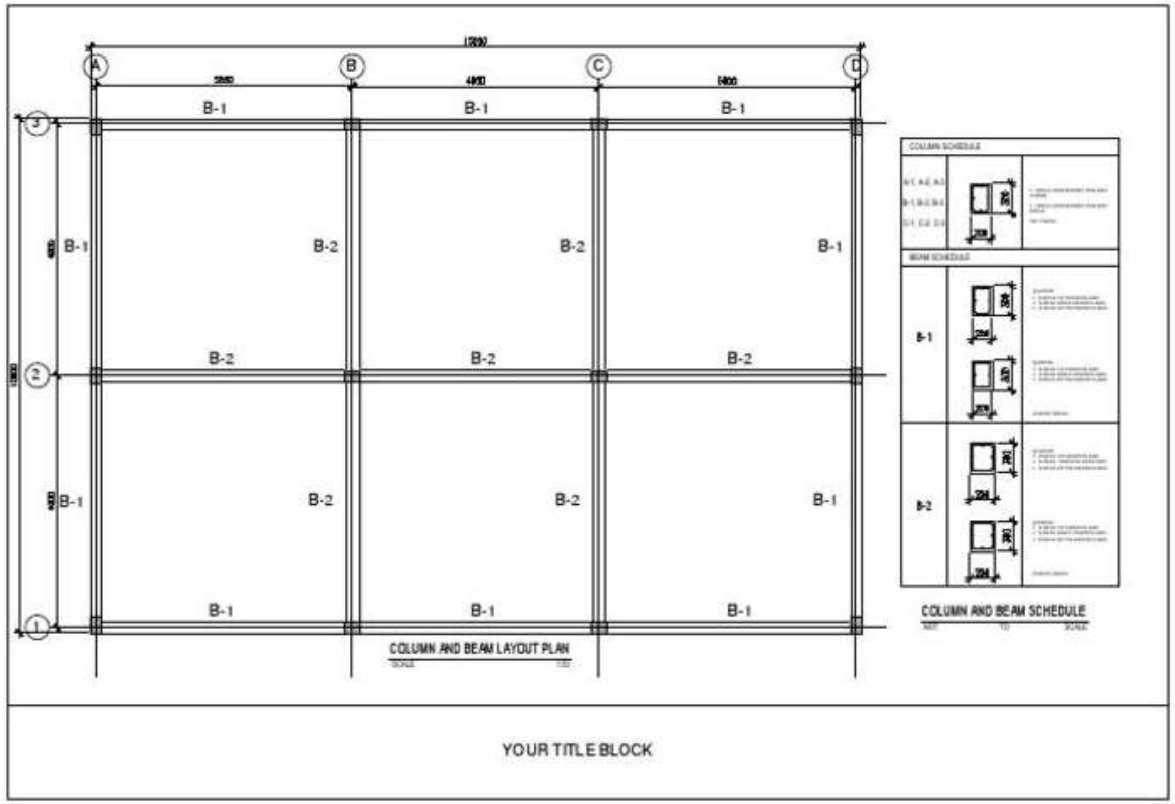
(a)



(b)

2. Civil engineering

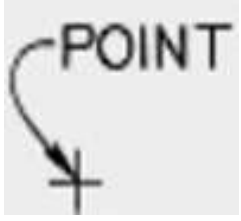
- Applications include:
  - ✓ Setting out perpendicular and parallel building lines.
  - ✓ Establishing geometric layouts for roads and site plans.
  - ✓ Constructing circles, arcs, and curves used in alignment drawings.
  - ✓ Preparing structural layouts and component details.
  - ✓ Dividing areas and locating reference points.

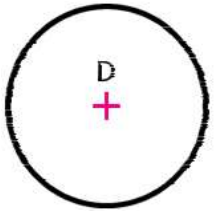


Basic Geometry

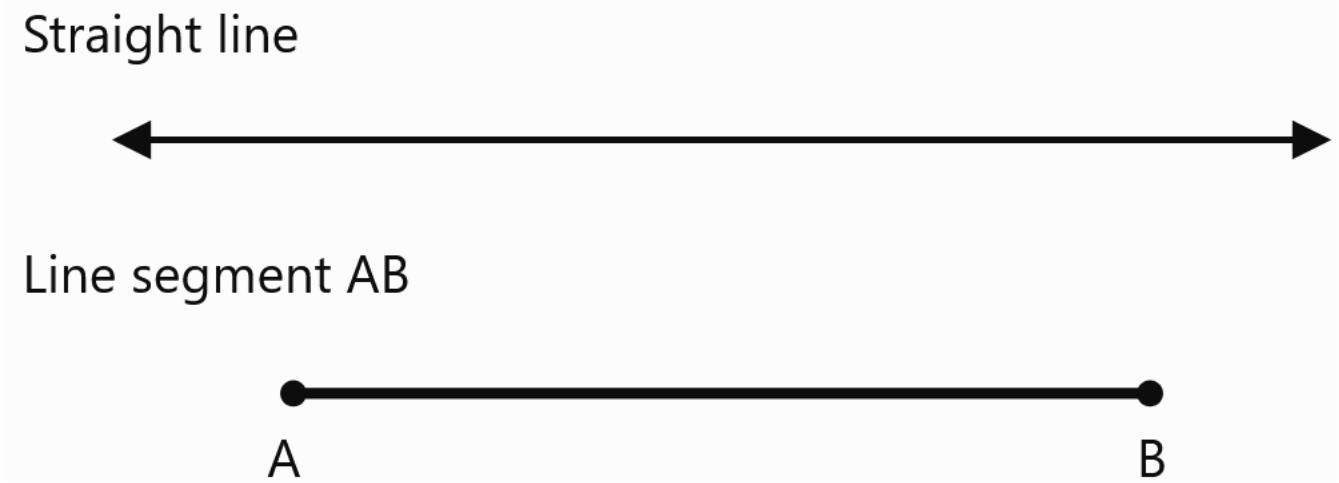
- Understanding basic geometric terminology is essential before beginning practical construction exercises. These terms describe the elements, relationships, and properties used to create geometric figures.
- The basic 2D geometric primitives from which other more complex geometric forms are derived. : Points, Lines, Circles, and Arcs.

1. **Point:** A point represents an exact position or location in space. It has no length, width, or thickness.
   - ✓ A point is normally identified by a capital letter, such as A, B, or C.
   - ✓ A point is represented in technical drawing as a small cross made of dashes that are approximately 3 mm long.
   - ✓ Its application is for locations of holes, centers, reference points, and intersections.

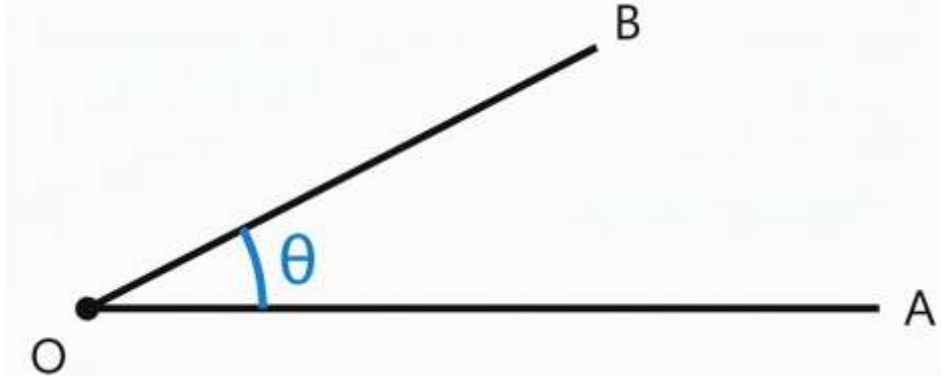




2. **Line:** A line is a straight geometric element extending continuously in a particular direction. A finite portion of a line is called a line segment.

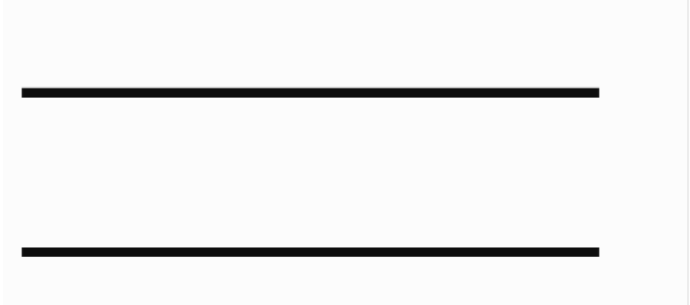


3. **Angle and vertex:** An angle is formed by two rays or line segments that share a common endpoint. The common endpoint is called the vertex. Angles are commonly measured in degrees (°).

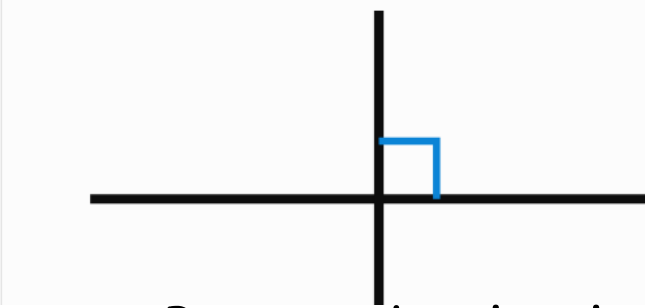


Angle AOB, with O as the vertex.

4. Parallel and perpendicular lines: Parallel lines are coplanar straight lines that never intersect and remain the same distance apart. Perpendicular lines intersect at exactly 90°.



Parallel lines



Perpendicular lines

5. Bisector and midpoint:

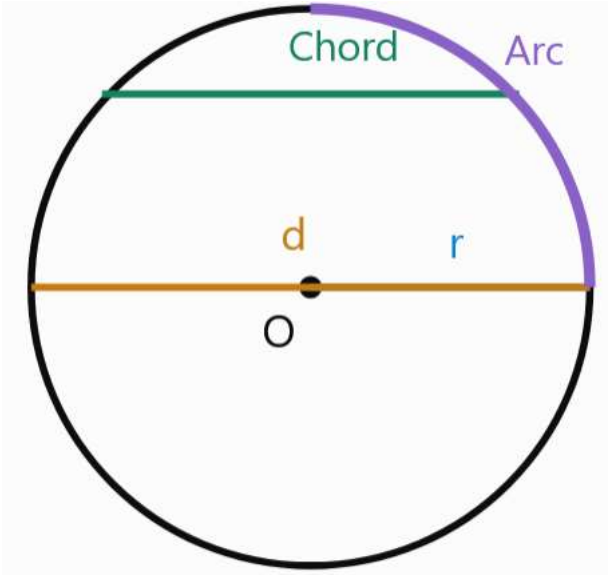
- ✓ Bisector: A line, ray, or construction that divides an angle or line segment into two equal parts.
- ✓ Midpoint: The point that divides a line segment into two equal lengths. If M is the midpoint of AB,

$$AM = MB = \frac{AB}{2}$$

6. Circle, arc, chord, radius, and diameter:

- A circle is a closed plane curve whose points are all at the same distance from a fixed point called the center.

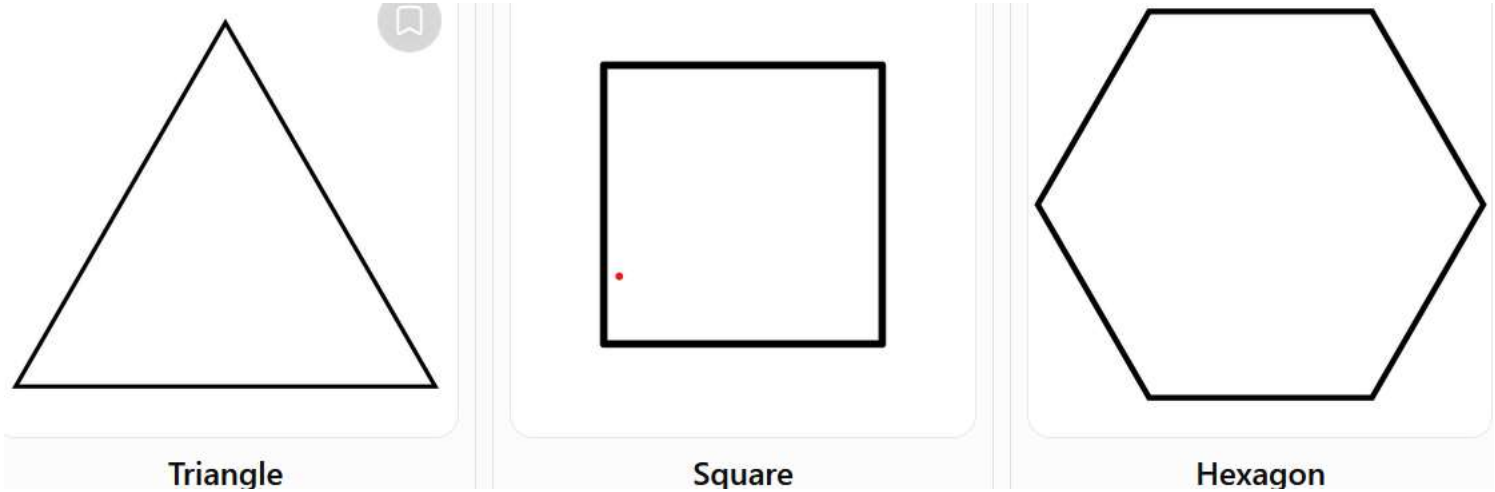
  - ✓ **Center:** The fixed point equidistant from every point on the circle.
  - ✓ **Radius (r):** A line segment from the center to any point on the circle.
  - ✓ **Diameter (d):** A chord passing through the center. It is twice the radius.
  - ✓ **Chord:** A line segment joining any two points on the circle.
  - ✓ **Arc:** A portion of the circumference of a circle.



Circle with center O

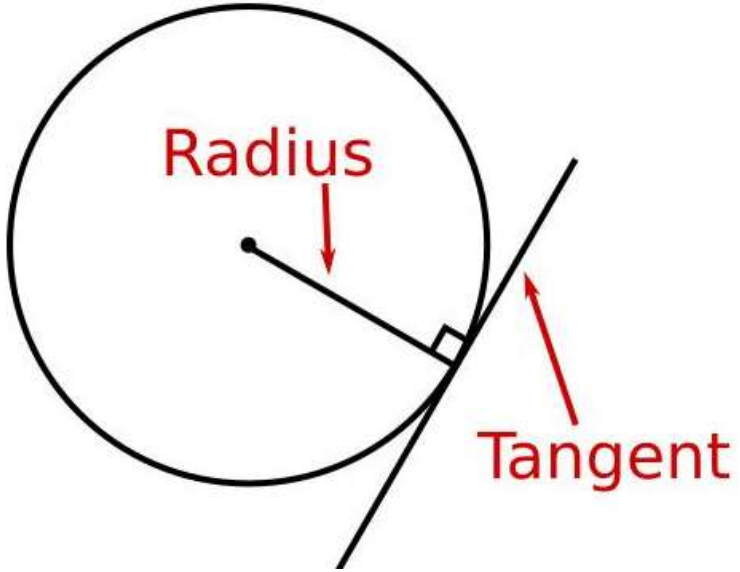
7. **Polygon:** A polygon is a closed plane figure bounded by three or more straight line segments. Each segment is a side, and each meeting point is a vertex. Polygons may be classified as regular or irregular.

- ✓ Regular polygon: All sides and all interior angles are equal.
- ✓ Irregular polygon: Its sides or interior angles are not all equal.



- ✓ The sum of the interior angles of an n-sided polygon is, $S = (n-2) \times 180^\circ$
- ✓ For a regular polygon, each interior angle is, $\alpha = \frac{(n-2)\times 180^\circ}{n}$

8. **Tangent:** A tangent is a straight line that touches a circle at exactly one point. For a circle, the tangent is perpendicular to the radius at the point of contact.



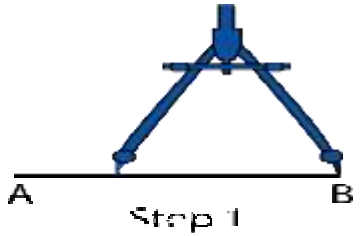
Geometric Construction Principles

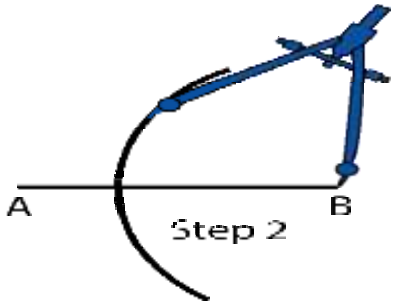
- Geometric construction follows established rules that help students create accurate figures using drawing instruments.

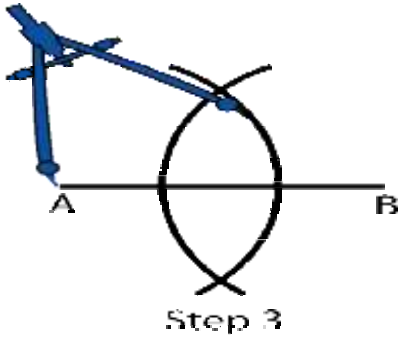
Bisecting a line

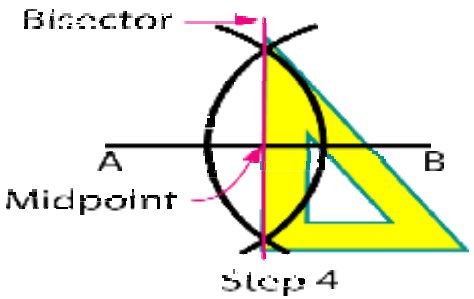
- With A as the centre and a radius greater than half AB, draw two arcs. With B as the centre and with the same radius, draw another arc intersecting the previous arcs. The line joining the intersection points is the perpendicular bisector of the line AB.

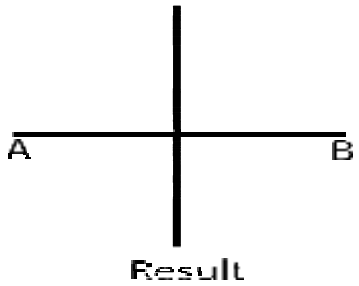




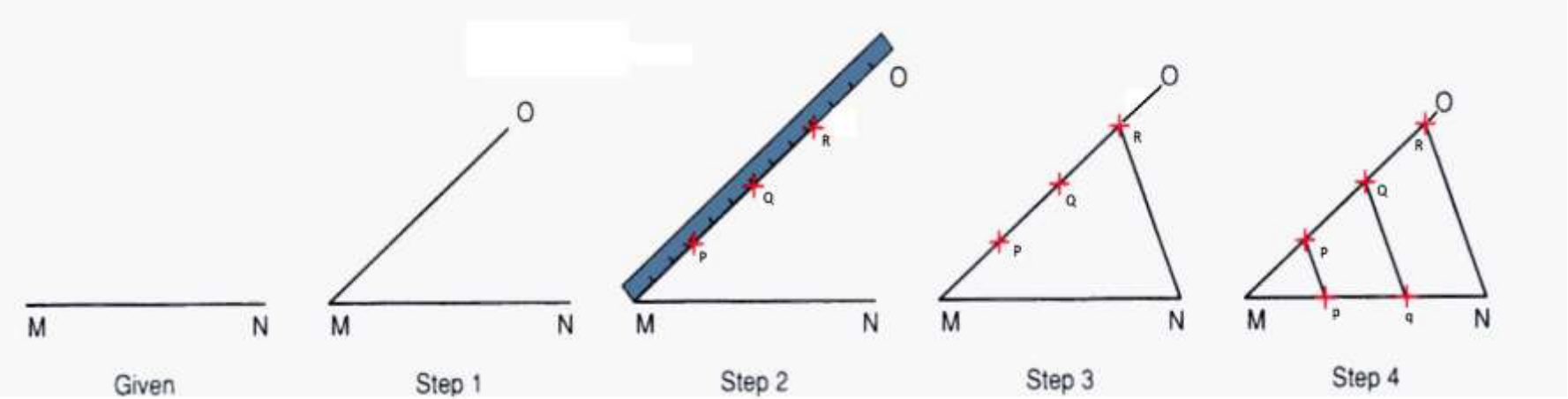






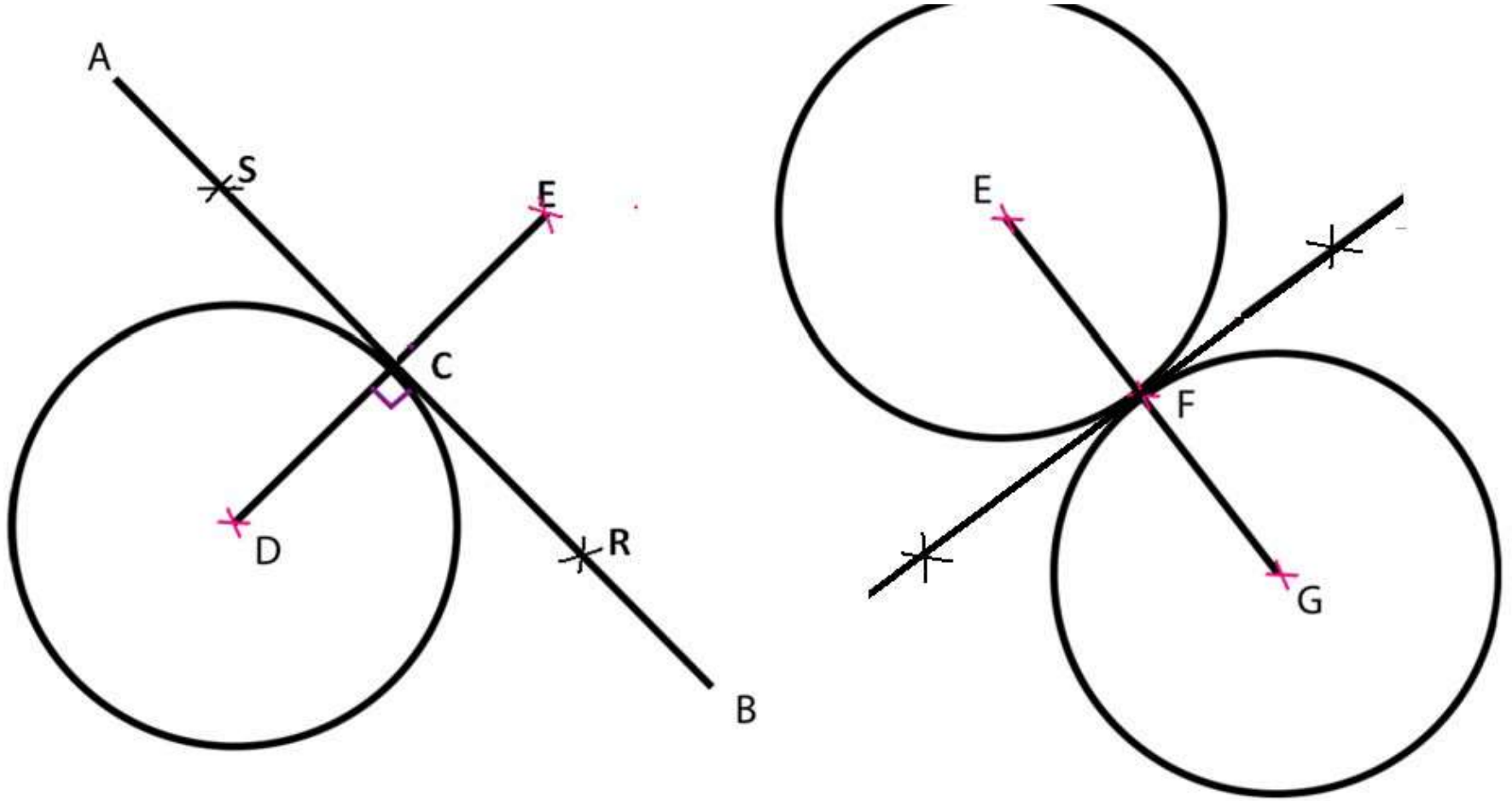


Dividing a line into equal parts



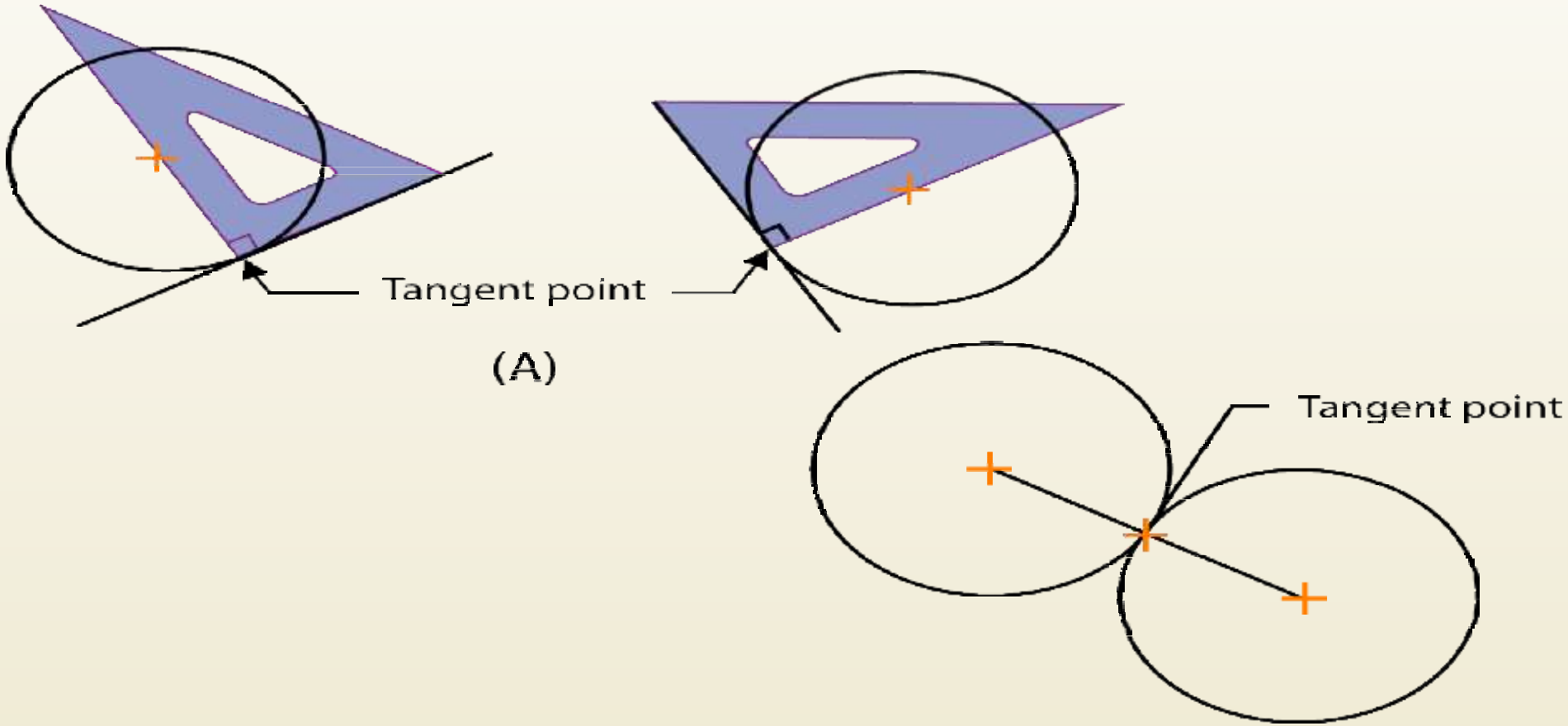
- Draw a line MO at any convenient angle (preferably an acute angle) from point M.
- From M and along MO, cut off with a divider equal divisions (say three) of any convenient length.
- Draw a line joining RN.
- Draw lines parallel to RN through the remaining points on line MO. The intersection of these lines with line MN will divide the line into (three) equal parts.

- *A planar tangent condition exists when two geometric forms meet at a single point and do not intersect.*

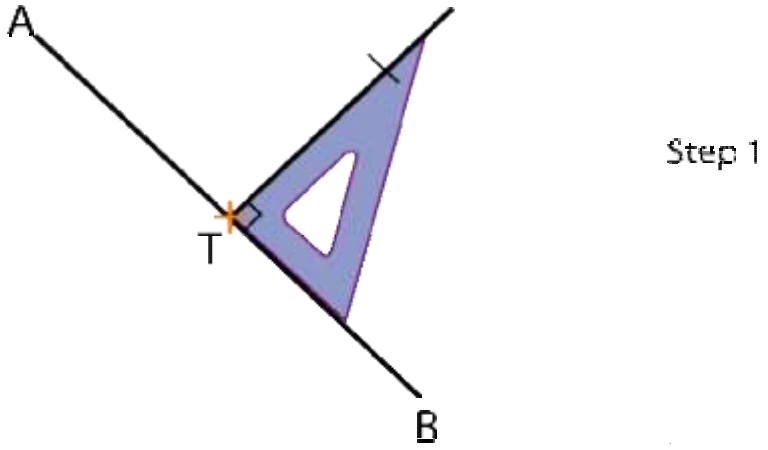


Locating tangent points on a circle and arcs

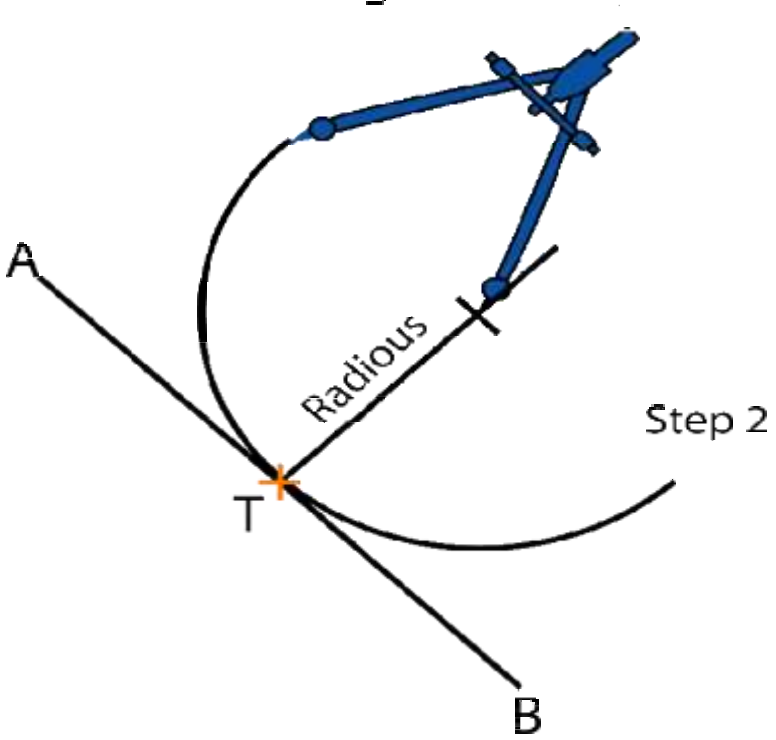
- The method of locating tangent points on a circle and arcs as well as the common tangent to two circles is shown in figures (A) and (B) .



Drawing an arc tangent to a given point on the line



Step 1



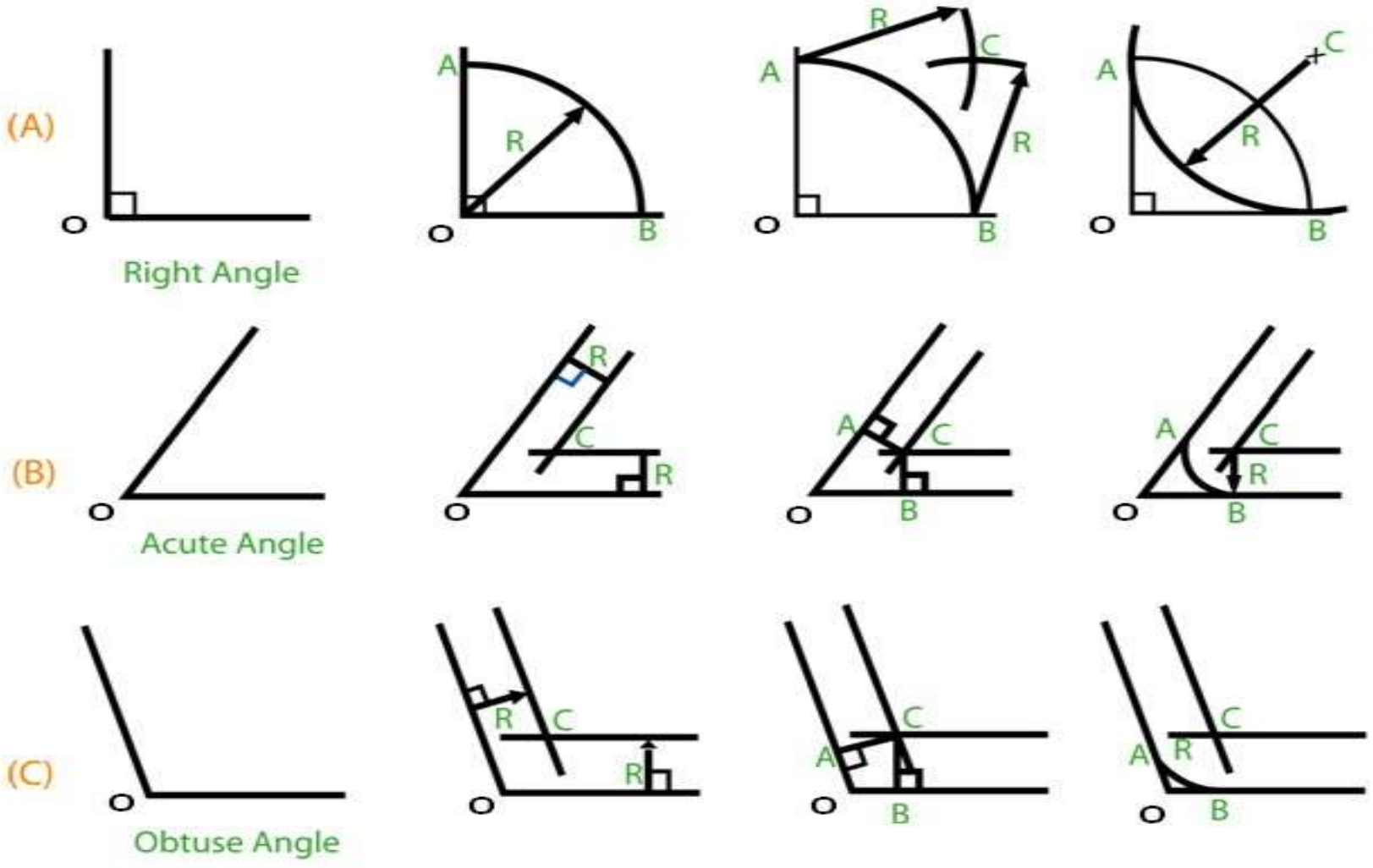
Step 2

Steps

- The steps for drawing the arc tangent to a given point on a line is shown as:
  1. Given line AB and tangent point T. Construct a line perpendicular to line AB and through point T.
  2. Locate the center of the arc by making the radius on the perpendicular line. Put the point of the compass at the center of the arc, set the compass for the radius of the arc, and draw the arc which will be tangent to the line through the point T.

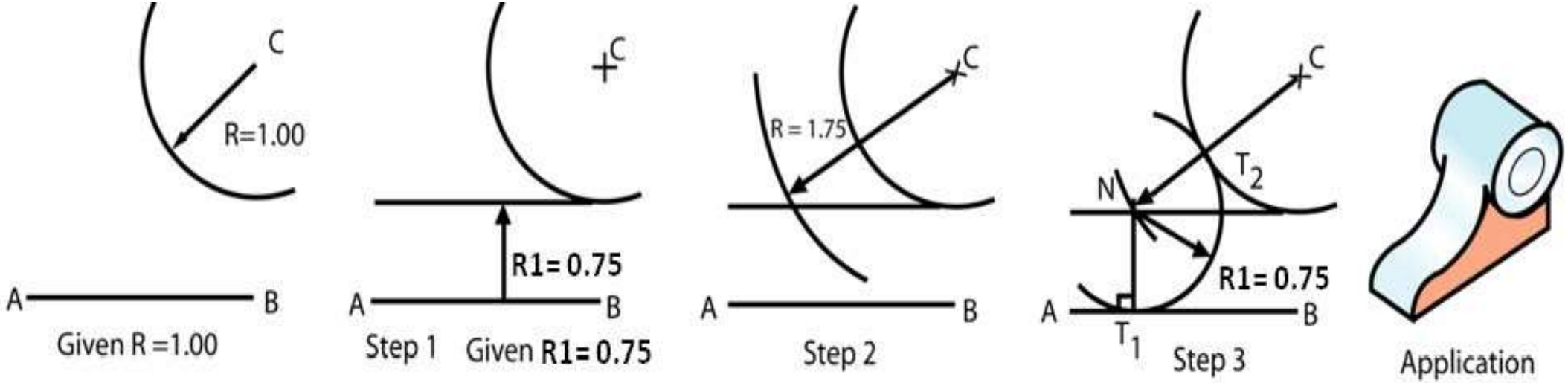
Drawing an arc, tangent to two lines

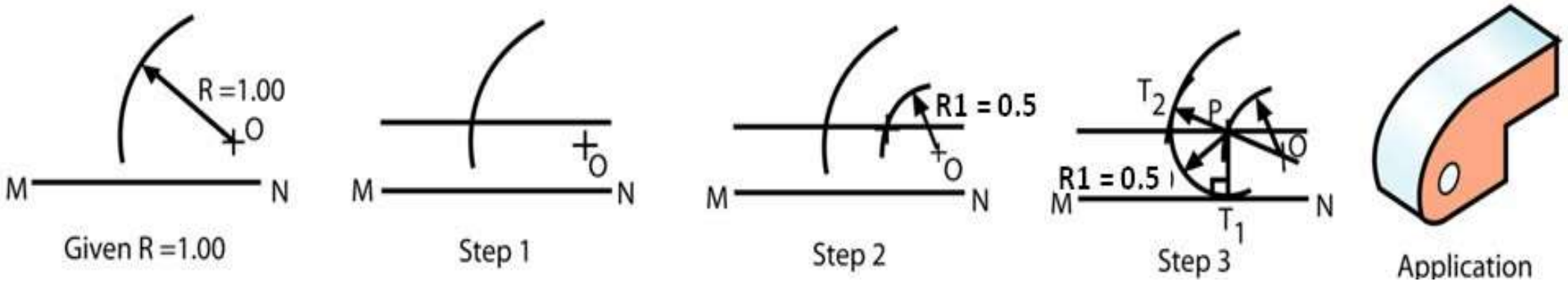
- The steps used to draw an arc tangent to two lines:



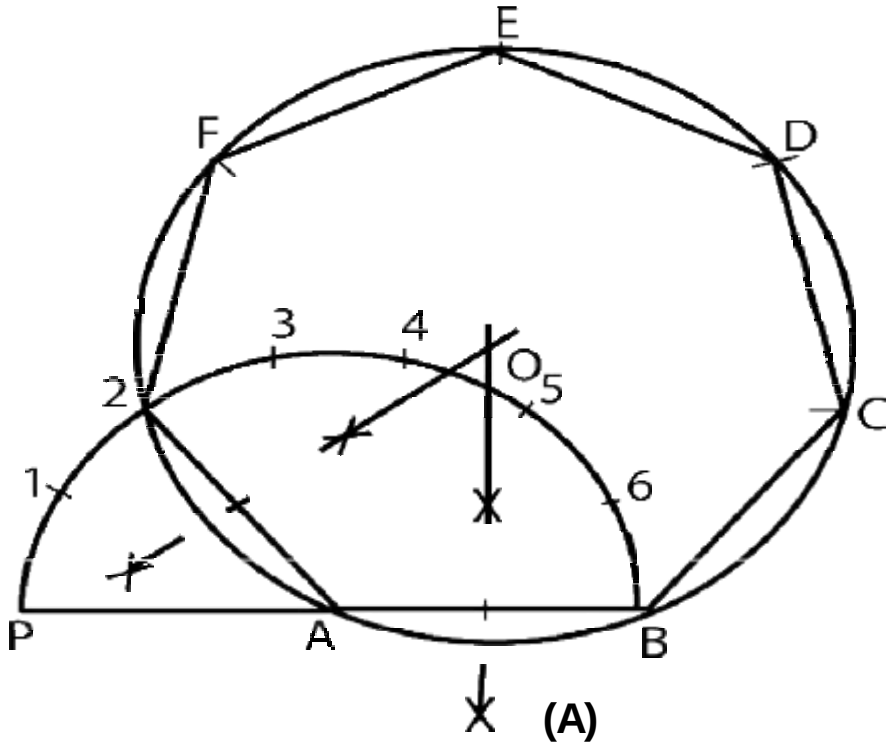
Drawing an arc, tangent to a line and an arc

- The steps in drawing an arc tangent to a line and an arc that (a) do not intersect (b) intersect each other.



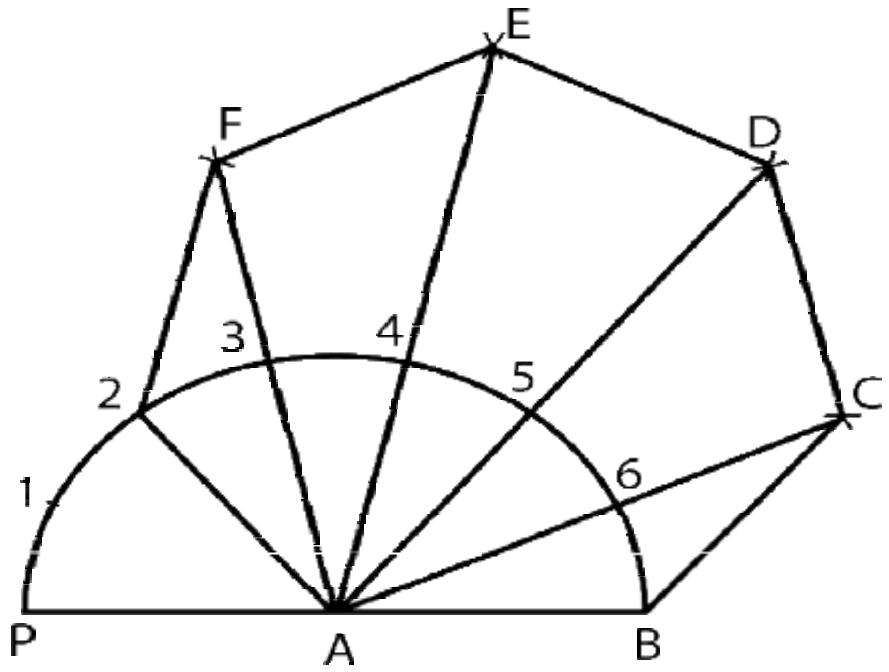


Construction of Regular Polygon of given length AB



(A)

Draw a line of length AB.

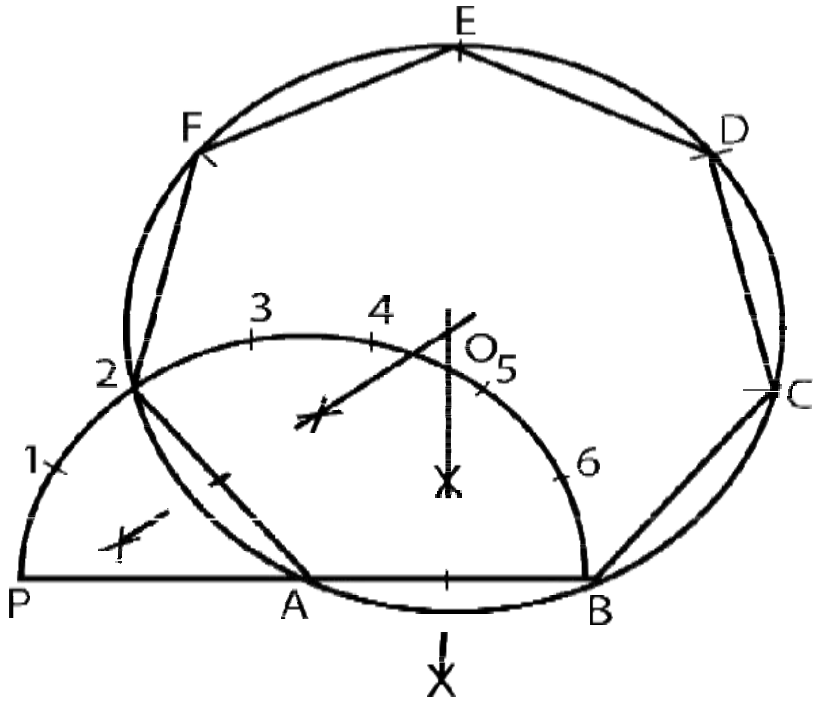


(B)

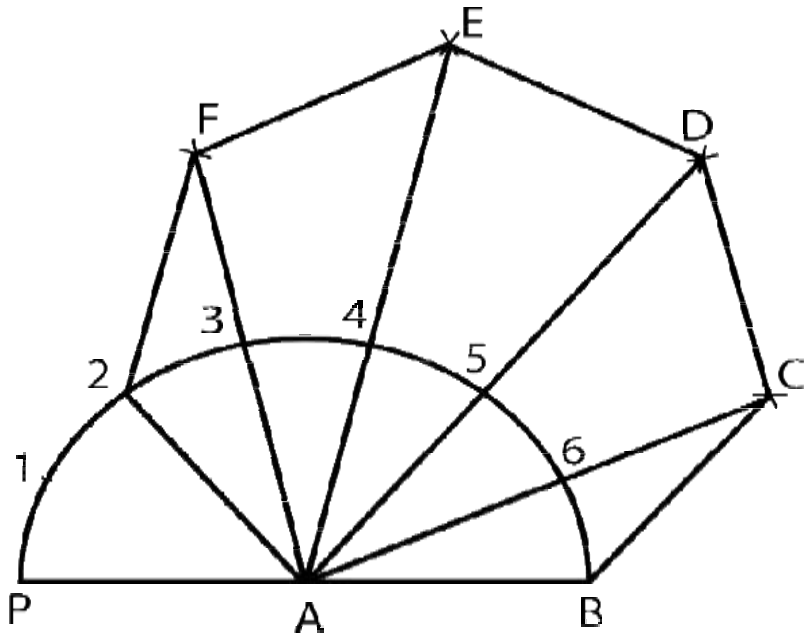
With A as centre and radius AB, draw a semicircle.

- With the divider, divide the semicircle into the number of sides of the polygon.
- Draw a line joining A with the second division-point 2.

Construction of Regular Polygon of given length AB…..



(A)

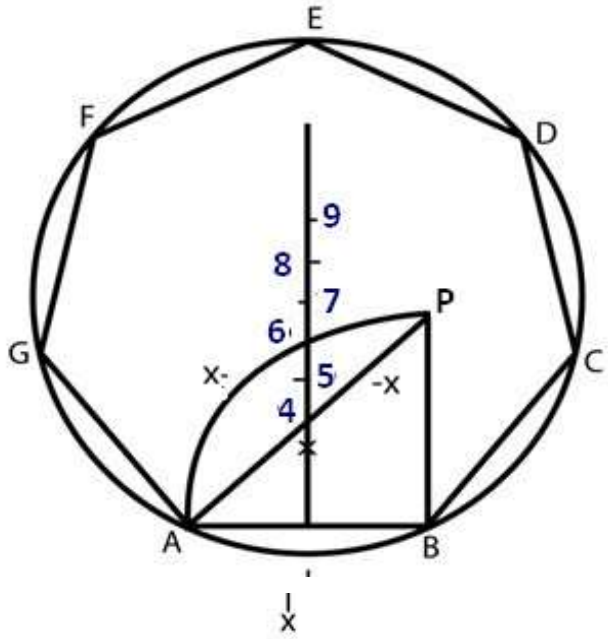


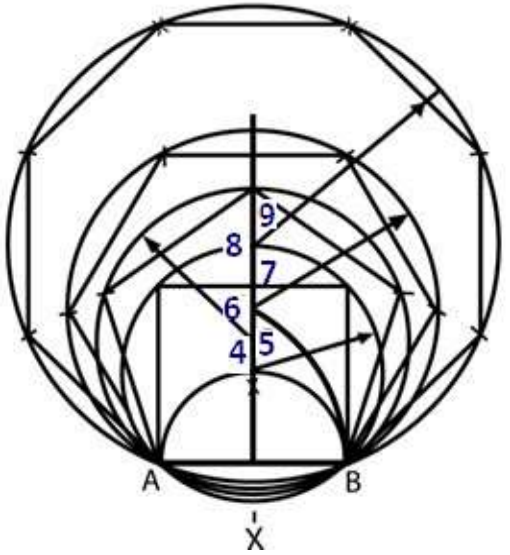
(B)

- The perpendicular bisectors of A2 and AB meet at O. Draw a circle with centre O and radius OA. With length A2, mark points F, E, D, and C on the circumference starting from 2 (*Inscribe circle method*)
- With centre B and radius AB draw an arc cutting the line A6 produced at C. Repeat this for other points D, E & F (*Arc method*)

General method of drawing any polygon

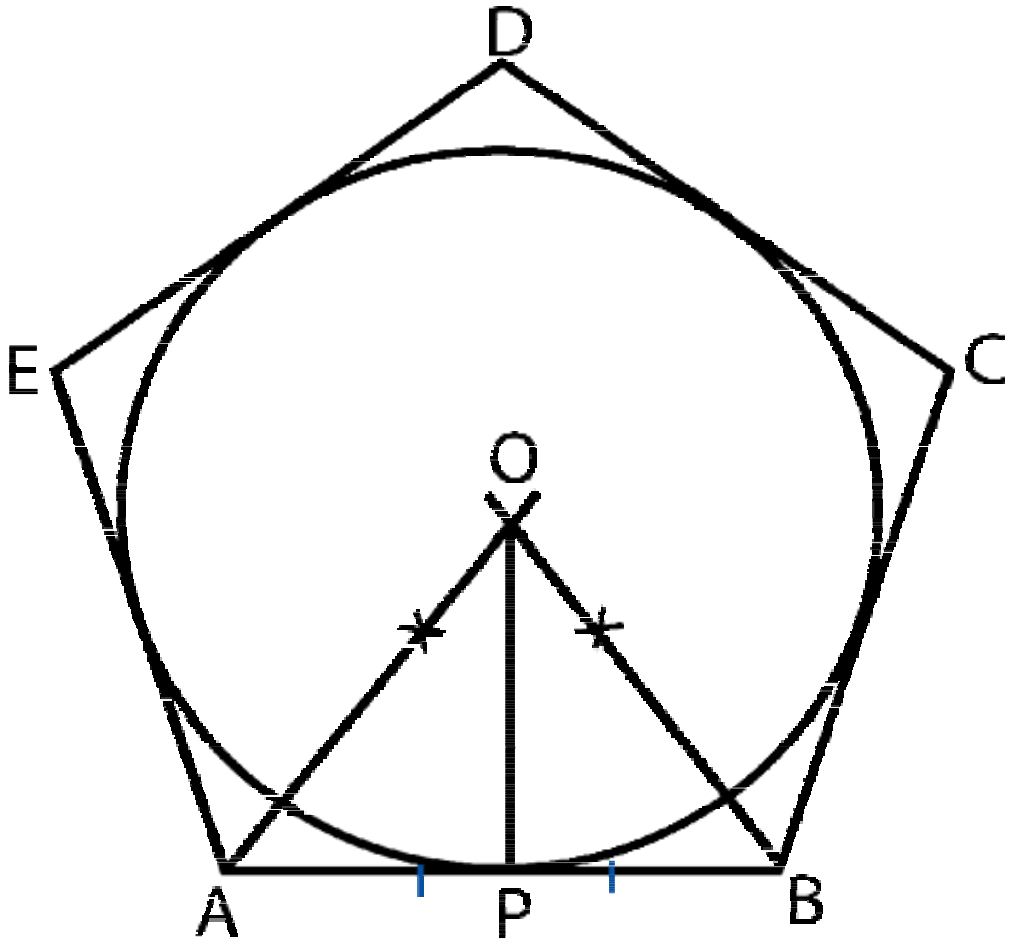
- Draw AB = the given length of the polygon. At B, draw BP perpendicular to AB
- Draw a straight line AP
- With center B and radius AB, draw arc AP.
- The perpendicular bisector of AB meets st. line AP and arc AP at 4 and 6, respectively.
- Draw circles with centers at 4, 5 and 6 and radii as 4B, 5B, and 6B and inscribe a square, pentagon, & hexagon in the respective circles.
- Mark points 7, 8, etc with 6-7,7-8, etc. = 4-5 to get the centers of the circles of the heptagon and octagon, etc.





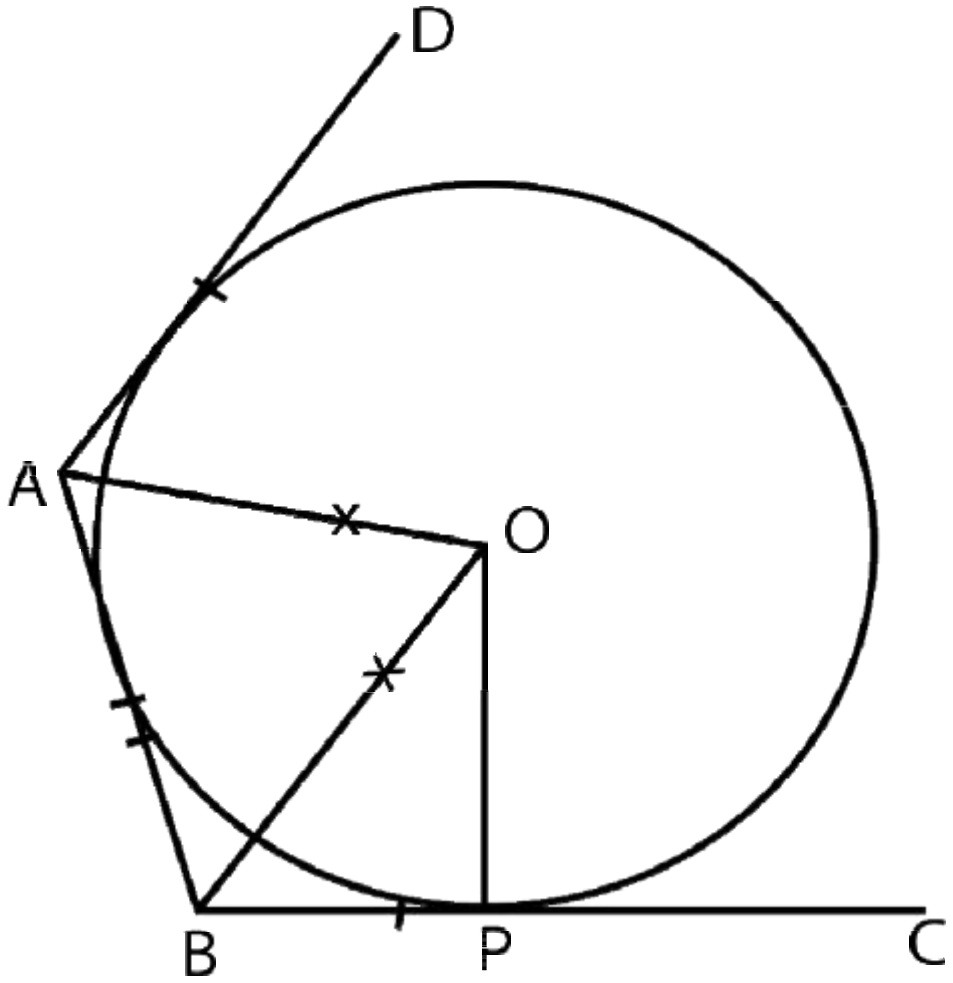
Inscribe a circle inside a regular polygon

- Bisect any two adjacent internal angles of the polygon.
- From the intersection of these lines, draw a perpendicular to any one side of the polygon (say OP).
- With OP as the radius, draw the circle with O as the center.



- To draw a circle touching three lines inclined to each other but not forming a triangle.

1. Let AB, BC, and AD be the lines.
2. Draw the bisectors of the two angles, intersecting at O.
3. From O, draw a perpendicular to any one line, intersecting it at P. With O as the center and OP as the radius, draw the desired circle.

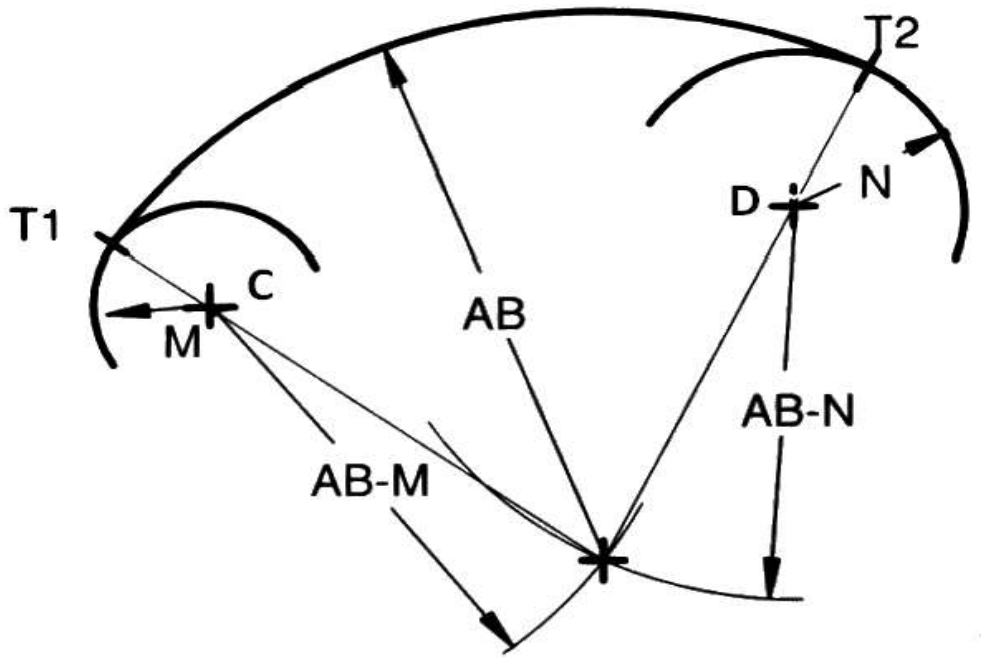


- **Construction of an arc tangent of given radius to two given arcs:**

➢ Given: Arcs of radii M and N. Draw an arc of radius AB units that is tangent to both the given arcs. The centers of the given arcs lie inside the required tangent arc.

Steps:

1. From centers C and D of the given arcs, draw construction arcs of radii (AB – M) and (AB N), respectively.
2. With the intersection point as the center, draw an arc of radius AB.
3. This arc will be tangent to the two given arcs.
4. Locate the tangent points T1 and T2.

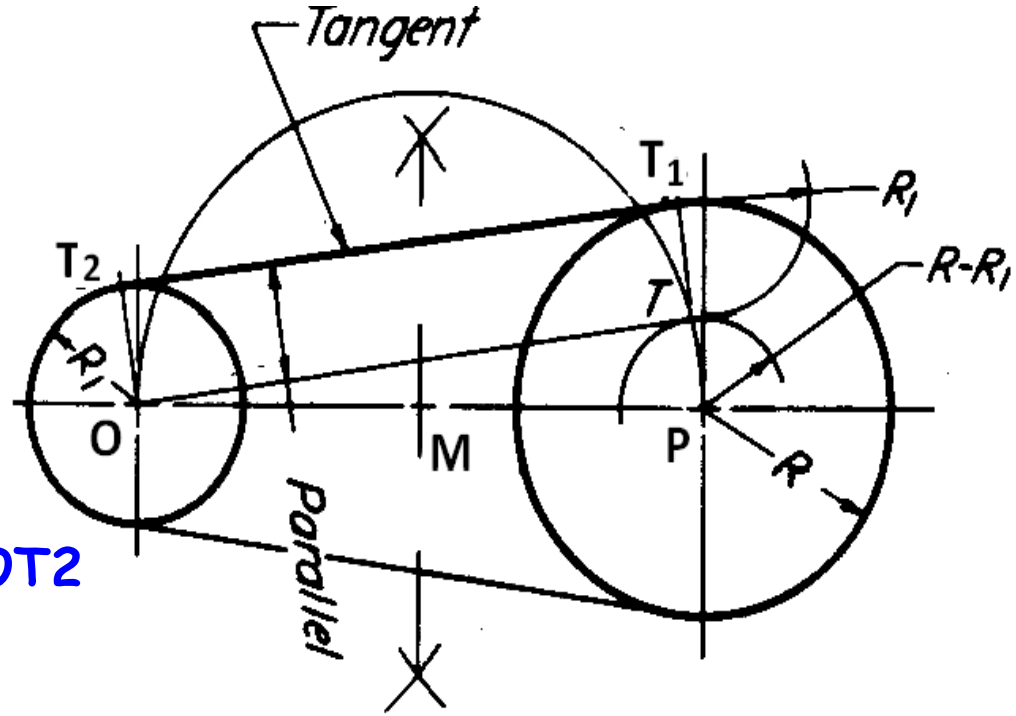


- **Construction of line tangents to two circles (Open belt)**

Given: Circles of radii R1 and R with centers O and P, respectively.

Steps:

- ✓ With P as center and a radius equal to **(R-R1)** draw an arc.
- ✓ Locate the midpoint of **OP as** the perpendicular bisector of **OP** as "**M**".
- ✓ With as centre and as radius, draw a semicircle.
- ✓ Locate the intersection point **T** between the semicircle and the circle with radius (**R-R1**).
- ✓ Draw a line **PT** and extend it to locate **T1**. Draw **OT2** parallel to **PT1**.
- ✓ The line **T1** to **T2** is the required tangent

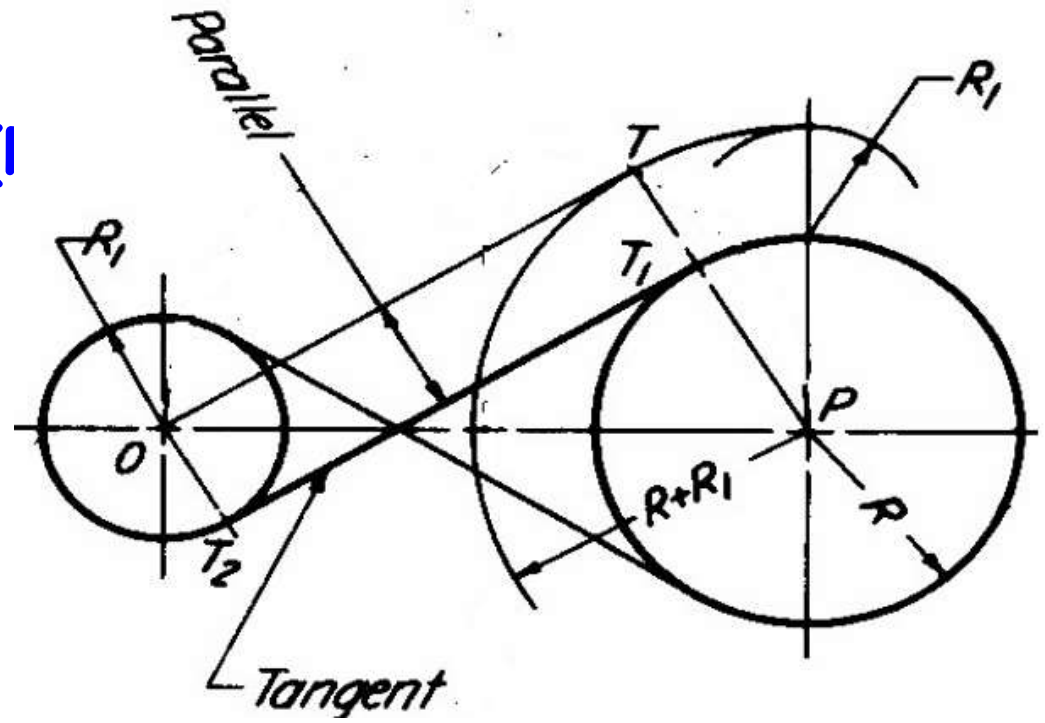


Construction of line tangents to two circles (crossed belt)

- **Given: Two circles of radii R1 and R with centers O and P, respectively.**

Steps:

1. Using **P** as a center and a radius equal to **(R + R1)**, draw an arc.
2. Through **O**, draw a tangent to this arc.
3. Draw a line PT cutting the circle at T_1
4. Through O draw a line OT_2 parallel to PT_1.
5. The line T_1T_2 is the required tangent.



End of Chapter 1